
TECHNISCHE UNIVERSITÄT BERLIN

Intelligent Multi-Robot Coordination Lab

Master's Thesis

Motion Planning and Control for Fully Submerged Hydrofoil Vessels

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Abstract

Hydrofoil vessels are emerging as a promising solution for sustainable, high-speed waterborne transportation. By reducing hydrodynamic drag through hull elevation, they enable energy-efficient operation. Fully submerged hydrofoils offer superior performance compared to surface-piercing designs but introduce significant challenges for control. They have highly nonlinear dynamics, six degrees of freedom, and strict constraints for stability. These characteristics demand advanced motion planning and control strategies capable of operating in dynamic environments.

This thesis investigates motion planning and control through the development of a novel dynamic model of a fully submerged hydrofoil vessel, qualitatively validated using a physical scale model. On this foundation, sampling-based motion planning algorithms, including Rapidly-exploring Random Trees (RRT) and Stable Sparse-RRT (SST), are studied for their ability to generate feasible trajectories under hydrofoil-specific dynamics and obstacle constraints. For trajectory-following of the planned motion, full-state feedback control methods are explored, focusing on Linear Quadratic Regulator (LQR) and Model Predictive Control (MPC).

Zusammenfassung

Tragflügelboote entwickeln sich zu einer vielversprechenden Lösung für einen nachhaltigen, schnellen Transport auf dem Wasserweg. Das Anheben des Rumpfes aus dem Wasser reduziert den hydrodynamischen Widerstand und ermöglicht dadurch einen sehr energieeffizienten Betrieb. Boote mit volleingetauchten Tragflügeln in T-Form bieten im Vergleich zu teileingetauchten Tragflügeln einen großen Effizienzvorteil, stellen jedoch erhebliche Herausforderungen an die Steuerung. Dieser Bootstyp weist eine stark nichtlineare Dynamik, sechs Freiheitsgrade und einen schmalen Betriebsbereich auf. Diese Eigenschaften erfordern fortschrittliche Bahnplanungs- und Steuerungsstrategien, die in dynamischen Umgebungen eingesetzt werden können.

Diese Arbeit beschäftigt sich mit der Entwicklung eines neuartigen dynamischen Modells für die Bahnplanung und -steuerung eines Tragflügelboots mit vollständig untergetauchten Tragflügeln. Das dynamische Modell wird anhand eines physikalischen Maßstabsmodells qualitativ validiert. Auf dieser Grundlage werden samplingbasierte Bahnplanungsalgorithmen – insbesondere Rapidly-exploring Random Trees (RRT) und Stable Sparse-RRT (SST) – daraufhin untersucht, inwieweit sie unter Berücksichtigung der tragflügelbootspezifischen Dynamik und bestehender Hindernisbeschränkungen realisierbare Trajektorien generieren können. Für die Trajektorienverfolgung der geplanten Bahn werden verschiedene rückgekoppelte Zustandsregler geprüft, wobei der Schwerpunkt auf dem Ricatti-Regler (LQR) und der Model Prädiktive Regelung (MPC) liegt.

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Nomenclature

Symbol	Description	Dimension	Unit
t	time	$\mathbb{R}$	s
x	state vector	$\mathcal{X}$	-
u	input vector	$\mathcal{U}$	-
f	dynamics function	$\mathcal{X} \times \mathcal{U} \rightarrow \mathcal{X}$	-
y	measurements	$\mathcal{Y}$	-
h	measurement function	$\mathcal{X} \times \mathcal{U} \rightarrow \mathcal{Y}$	-
$\mathcal{B}$	fixed body reference frame	-	-
$\mathcal{W}$	fixed world reference frame	-	-
p	position	$\mathbb{R}^3$	m
x	x-position	$\mathbb{R}$	m
y	y-position	$\mathbb{R}$	m
z	z-position	$\mathbb{R}$	m
R	orientation matrix	$\mathbb{R}^{3 \times 3}$	-
q	orientation unit quaternion	$\mathbb{H}$	-
φ	roll angle	$\mathbb{R}$	rad
θ	pitch angle	$\mathbb{R}$	rad
ψ	yaw angle	$\mathbb{R}$	rad
v	linear velocity	$\mathbb{R}^3$	m s^{-1}
u	linear velocity in x	$\mathbb{R}$	m s^{-1}
v	linear velocity in v	$\mathbb{R}$	m s^{-1}
w	linear velocity in w	$\mathbb{R}$	m s^{-1}
ω	angular velocity	$\mathbb{R}^3$	rad s^{-1}
p	roll rate	$\mathbb{R}$	rad s^{-1}
q	pitch rate	$\mathbb{R}$	rad s^{-1}
r	yaw rate	$\mathbb{R}$	rad s^{-1}
$\mathcal{X}$	state space / configuration space	-	-
g	gravity vector	$\mathbb{R}^3$	m s^{-2}
m	vessel mass	$\mathbb{R}$	kg
J	inertia matrix	$\mathbb{R}^{3 \times 3}$	kg m^2
F	total force vector	$\mathbb{R}^3$	N
τ	total moment vector	$\mathbb{R}^3$	N m
$\mathcal{F}$	flow velocity reference frame	-	-
V_a	advance velocity	$\mathbb{R}$	m s^{-1}

β	sideslip angle	$\mathbb{R}$	rad
α	angle of attack	$\mathbb{R}$	rad
q	dynamic pressure	$\mathbb{R}$	N m^{-2}
S	reference area	$\mathbb{R}$	m^2
b	wingspan	$\mathbb{R}$	m
c	chord length	$\mathbb{R}$	m
ρ	fluid density	$\mathbb{R}$	kg m^{-3}
P	power	$\mathbb{R}$	W
u_{thrust}	thrust input	$\mathbb{R}$	W
F_{thrust}	thrust force	$\mathbb{R}$	N
$\mathbf{u}$	input vector	$\mathcal{U}$	-
u_{stb}	starboard foil pitch input	$\mathbb{R}$	$^\circ$
u_{prt}	port foil pitch input	$\mathbb{R}$	$^\circ$
u_{stn}	stern foil pitch input	$\mathbb{R}$	$^\circ$
u_{rud}	rudder angle input	$\mathbb{R}$	$^\circ$
$\mathcal{X}_{\text{obs}}$	obstructed configurations space	-	-
$\mathcal{X}_{\text{free}}$	free configuration space	-	-
τ	path	$\mathbb{R} \rightarrow \mathcal{X}$	-
$\mathbf{x}_s$	start configuration	$\mathcal{X}$	-
$\mathbf{x}_g$	goal configuration	$\mathcal{X}$	-
δ_{BN}	SST selection parameter	$\mathbb{R}$	-
δ_s	SST pruning parameter	$\mathbb{R}$	-
J	cost function	-	-
$\mathbf{A}$	dynamics matrix	$\mathbb{R}^{n \times n}$	-
$\mathbf{B}$	input matrix	$\mathbb{R}^{n \times p}$	-
$\mathbf{C}$	output matrix	$\mathbb{R}^{q \times n}$	-
$\mathbf{D}$	feed trough matrix	$\mathbb{R}^{q \times p}$	-
π	policy	-	-
$\mathbf{Q}$	state weight matrices	$\mathbb{R}^{n \times n}$	-
$\mathbf{R}$	input weight matrices	$\mathbb{R}^{p \times p}$	-
$\mathbf{K}$	control matrix	$\mathbb{R}^{p \times n}$	-
$\mathbf{p}_i^{\mathcal{B}}$	position of foil i	$\mathbb{R}^3$	m
$\mathbf{q}_i^{\mathcal{B}}$	orientation of foil i	$\mathbb{H}$	-
c_i	chord length of foil i	$\mathbb{R}$	m
S_i	reference area of foil i	$\mathbb{R}$	m^2
γ_i	pitch angle of foil i	$\mathbb{R}$	rad

α_i	angle of attack	$\mathbb{R}$	rad
F_L	lift force	$\mathbb{R}$	N
F_D	drag force	$\mathbb{R}$	N
Re	Reynolds number	$\mathbb{R}$	-
ν	waters kinematic viscosity	$\mathbb{R}$	N s m^{-2}
$\mathcal{F}_i$	foil i fixed reference frame	-	-

List of Acronyms

Acronym	Description
DOF	Degrees of Freedom
ENU	East-North-Up Coordinates
GPS	Global Positioning System
IMU	Interial Mearument Unit
IOPT	Interior Point Optimizer
LQR	Linear Quadratic Regulator
LQRI	Linear Quadratic Regulator with Integral Action
LTI	Linear Time-Invariant Systems
MEBC	Monaco Energy Boat Challenge
MPC	Model Predictive Control
NMPC	Non-linear Model Predictive Control
MRP	Modified Rodrigues Parameters
NED	North-East-Down Coordinates
OMPL	Open Motion Planning Library
PID	Proportional-integral-derivative controller
PRM	Probabilistic Roadmap
RRT	Randomly Exploring Random Tree
SO	Special Orthogonal Group
SST	Stable Sparse RRT
UAV	Unmanned Aerial Vehicle
USV	Unmanned Survace Vehicle
VLM	Vortec Lattice Method
IPOPT	Interior Point OPTImizer

1. Introduction

In recent years, hydrofoil technology has gained renewed attention as a promising solution for high-speed and energy-efficient waterborne transportation. By lifting the vessel's hull above the water surface, hydrofoils significantly reduce hydrodynamic drag, enabling greater speeds while lowering energy consumption.

This technology has brought the greatest recent advancements with regards to high performance vessels such as sailing vessels seen in major competitions, such as Sail GP [1] or Americas Cup [2] competitions as well as international solar boats challenges.

These advantages align with the maritime sector's urgent drive toward carbon-neutral propulsion and sustainable vessel design. Hydrofoiling is increasingly viewed not only as a means of enhancing performance but also as a pathway toward environmentally responsible marine mobility [3], [4].

Hydrofoil vessels can be categorized into surface-piercing and fully submerged configurations. While surface-piercing foils provide inherent passive stability, their efficiency and performance are limited at higher speeds and in rough sea states. Fully submerged foils offer superior hydrodynamic efficiency and ride quality but require active control to maintain stability. The challenge lies in counteracting nonlinear hydrodynamic effects and external disturbances, while also ensuring safety and system reliability. Robust control methods are therefore central to unlocking the full potential of fully submerged hydrofoil vessels.

Beyond stability, emerging applications demand that hydrofoil vessels navigate dynamically changing environments.

Consider a scenario where an obstacle suddenly appears in the vessel's path: real-time trajectory planning, coupled with reliable control execution, becomes critical for safety. While control design for hydrofoils has received considerable research attention, motion planning remains comparatively underexplored. In contrast, extensive work exists in domains such as aerial robotics (e. g. fixed wing UAVs [5], quad rotors [6]) and autonomous surface vessels, where planning methods have enabled agile and safe navigation in cluttered environments [7](p. 1f). With new competitions for autonomous carbon-neutral vessels [8] and application of hydrofoils to autonomous surface vessel [9], [10], motion planning for these new vehicles is gaining interest.

The dynamics of hydrofoiling vessels pose unique challenges especially for motion planning: More degrees of freedom than conventional surface vessels, more complex dynamics than quad rotors, and tighter state constraints than fixed wing UAVs especially regarding height stability.

This thesis builds on these insights, aiming to research and adapt state-of-the-art motion planning and control methods to the unique challenges posed by fully submerged hydrofoil vessels.

2. Background

The two main topics of the research are *Motion Planning* and *Control*. But every Planning and Control Algorithm relies on some kind of forward model that describes how the robot, vehicle or plant reacts to different inputs and interacts with the environment. Therefore, we must firstly discuss *System Dynamics* of hydrofoiling vessels as it forms the foundation for motion planning and control.

2.1. System Dynamics

This section describes the process of developing mathematical models defining the behavior of a dynamic system. The goal is to create a model that accurately represents how the system responds to different inputs. Depending on the available knowledge about the system, system identification can be approached using three different modeling techniques. *Black-box* modeling uses only observed input-output data, without any assumptions about the system's internal workings. *Grey-box* modeling combines some knowledge about the system with data-driven algorithms. By contrast, *White-box* modeling relies on a deep understanding of the system's internal structure, using physical laws to create the model from first principles. This is the technique we will mostly rely on for this research.

2.1.1. Hydrofoil Vessels

Hydrofoiling vessels are a class of marine vehicles that utilize submerged wings to generate lift and elevate the hull above the water surface, thereby reducing drag by decreasing the wetted area and increasing efficiency. Two primary approaches are employed: *Surface-piercing hydrofoils* extend above the waterline in a V-shape. This geometry leads to restoring forces back to an equilibrium position, causing stability in heave, roll and pitch in foilborne condition [11] (p. 166). The passive stability comes at the cost of hydrodynamic efficiency losses due to the surface piercing [12](p. 347). *Fully submerged hydrofoils* have no passive stability but allow more efficient lift and fewer perturbations due to waves. The foils feature actuator systems allowing to change the lift of each wing. Stability is achieved with active control algorithms. This setup is not only more efficient but also allows banking in turns, improving maneuverability, and has better seakeeping characteristics [12](p. 348), [11](p. 176, 178).



Figure 1: 1:3 Scale boat on a canal in Berlin.

2.1.2. Vessel Description

The examined vessel shown in Figure 1 is a twinhull with three fully submerged T-foils. Two narrow hulls on each side provide the necessary hydrostatic lift. The crossbeams between the hulls provide a mount for the foil setup. The three foils assemble a similar setup. They consist of a flow-optimized mast and a lifting foil. The lifting foil is mounted on a pivot, allowing for pitch actuation via a pushrod. Two inverted T-foils are mounted on the front crossbeam. A third inverted T-foil is mounted at the center of the aft crossbeam. The aft masts yaw can be actuated to act as a rudder. For thrust, an electric pod-motor is integrated into the aft mast. Changing the rudder angle rotates the thrust vector.

The vessel operates in three hydrodynamic modes. At slow speed the hulls are in displacement mode, where vessel is supported predominantly by the hydrostatic lift of the hulls. With increasing speed, the hulls develop dynamic pressure, decreasing the draft by lifting the vessel and therefore reducing hydrodynamic drag. The third operating mode of the vessel is reached when the forward speed generates sufficient positive hydrodynamic pressure on the lifting foils such that the hulls are lifted completely out of the water. This greatly decreases drag once again and defines the most efficient mode. This fully foiling mode imposes a challenging control problem since the steady state is instable in roll, pitch and ride height. For deriving the model we are predominantly interested in the foiling mode.

2.1.3. State-Space Representation

The dynamics model can be mathematically expressed either by showing states as distribution over a frequency band in a *frequency domain*, or how a state changes over time in a *time domain*. The changes of states are often represented in a *state space* model. The vector $x(t)$ denotes the states over time t . The differential equation system

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t)), & \mathbf{x}(t_0) &= \mathbf{x}_0 \\ \mathbf{y}(t) &= \mathbf{h}(\mathbf{x}(t), \mathbf{u}(t))\end{aligned}\tag{1}$$

describes the change $\dot{\mathbf{x}}(t)$ depending on the current state $\mathbf{x}(t)$ and input vector $\mathbf{u}(t)$ with the dynamics function $\mathbf{f}$. The vector $\mathbf{y}(t)$ denotes the measurements which are described by the measurement model $\mathbf{h}$.

2.1.4. Reference Frames

To define the state vector, a coordinate system must be defined. Marine vessel design, aerospace and robotics use different kinds of reference systems to define the state of a body. Let us define the fixed body reference system $\mathcal{B}$, where a position $\mathbf{p}$ is described by x pointing forward, y pointing to port³ and z upwards. The origin lies in the body center of mass. The orientation $\mathbf{R}$ can be described by the Euler angles roll φ around x , pitch θ around y and yaw ψ around z , or as unit quaternion $\mathbf{q}$. The linear velocities $\mathbf{v}$ are $[u, v, w]^4$ and angular velocities denote $\boldsymbol{\omega} = [p, q, r]$.

There are two variants for fixed world coordinate systems, which exist on the local tangent plan of the Earth's surface. While in the aerospace community the North-East-Down (NED) coordinates are common, we will define the fixed world coordinated $\mathcal{W}$ in East-North-Up (ENU) coordinates, which are more common for surface applications. In this system the east axis is labeled x , the north y and up is labeled z .

2.1.5. Defining the State

In the process of modeling the dynamics of foiling vessels the state is $\mathbf{x} \in \mathcal{X}$ in configuration space defined as $\mathcal{X}$. The vessel is a rigid body in three-dimensional space. It has a position $\mathbf{p}$ in 3D-Euclidian space $\mathbb{R}^3$, an orientation $\mathbf{q}$ in 3D-orientation-group $\text{SO}(3)$, described by a unit quaternion, a velocity $\mathbf{v}$ in 3D-Euclidian space $\mathbb{R}^3$ and an angular velocity $\boldsymbol{\omega}$ in 3D-Euclidian space $\mathbb{R}^3$. The components get combined into the state vector

$$\mathbf{x} = [\mathbf{p}, \mathbf{q}, \mathbf{v}, \boldsymbol{\omega}]^\top \in \mathcal{X}\tag{2}$$

with the configuration space defined as the compound space $\mathcal{X} = \mathbb{R}^3 \times \text{SO}(3) \times \mathbb{R}^3 \times \mathbb{R}^3$. The Reference frames and units are defined in Table 1.

³In shipping, the directions on a boat are referred to using the following nautical terms: starboard (right), port (left), stern (rear) and bow (front).

⁴The linear velocities are also referred to as surge u for longitudinal motion, sway v for side to side movement, and heave w for vertical motion

State	Space	Unit	Frame
Position $\mathbf{p}$	$\mathbb{R}^3$	m	$\mathcal{W}$
Orientation $\mathbf{q}$	SO3	rad	$\mathcal{W}$
Velocity $\mathbf{v}$	$\mathbb{R}^3$	$\frac{m}{s}$	$\mathcal{W}$
Angular Velocity $\boldsymbol{\omega}$	$\mathbb{R}^3$	rad s^{-1}	$\mathcal{B}$

Table 1: State representation parameter

2.1.6. Defining the Inputs

The input vector

$$\mathbf{u} = [u_{\text{stb}}, u_{\text{prt}}, u_{\text{stn}}, u_{\text{rud}}, u_{\text{thrust}}]^\top \in \mathcal{U} \subseteq \mathbb{R}^5 \quad (3)$$

is defined by the controllable inputs of the vessel. The first three entries u_{stb} , u_{prt} and u_{stn} describe the actuated pitch angle of the lifting foils depicted in Figure 2. u_{rud} is the actuation angle of the rudder. Finally u_{thrust} defines the forward thrust power acting on the vessel.

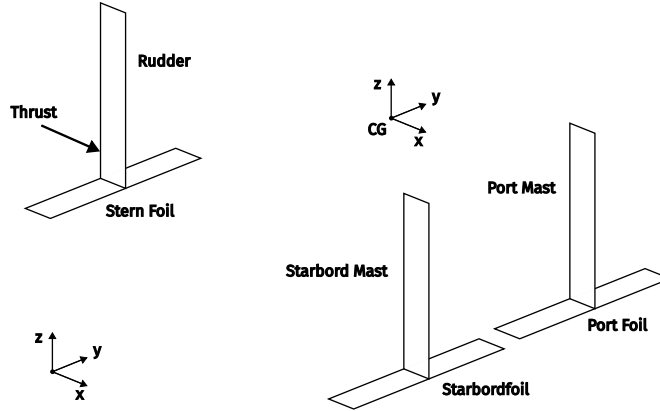


Figure 2: Schematic view of the hydrofoil vessel

2.1.7. Rigid Body Dynamics

Rigid body dynamics describe how solid objects move when forces and torques act on them, assuming the bodies do not deform. It combines linear and rotational motion to describe translation and rotation of objects in space. The dynamic function $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$ models the change of state regarding the inputs $\mathbf{u}$. With the gravity vector $\mathbf{g} \in \mathbb{R}^3$ facing downward in world space, the vessel mass $m \in \mathbb{R}$ and moment of inertia matrix $\mathbf{J} \in \mathbb{R}^{3 \times 3}$, the dynamics can be written as [13](p. 15):

$$\mathbf{f}(\mathbf{x}, \mathbf{u}) = \dot{\mathbf{x}} = \begin{pmatrix} \dot{\mathbf{p}} \\ \dot{\mathbf{q}} \\ \dot{\mathbf{v}} \\ \dot{\boldsymbol{\omega}} \end{pmatrix} = \begin{pmatrix} \mathbf{v} \\ \mathbf{q} \otimes (0, \boldsymbol{\omega})_{\mathbb{H}} \\ \mathbf{g} + \mathbf{q} \odot \left(\frac{1}{m} \mathbf{F}(\mathbf{x}, \mathbf{u}) \right) \\ \boldsymbol{\omega} \times \mathbf{J}^{-1} \boldsymbol{\omega} + \mathbf{J}^{-1} (\boldsymbol{\tau}(\mathbf{x}, \mathbf{u})) \end{pmatrix} \quad (4)$$

The functions $\mathbf{F}(\mathbf{x}, \mathbf{u})$ and $\boldsymbol{\tau}(\mathbf{x}, \mathbf{u})$ describe forces and moments applied to the vessel in fixed body coordinates. Rotation of a vector by a quaternion is defined the operator $\odot$ and the quaternion multiplication as $\otimes$.⁵

2.1.8. Modeling Hydrodynamic Forces

In this section different approaches are discussed to model the hydro dynamic forces generated by the foils interacting with water.

2.1.8.1. Airplane Flight Dynamics

A hydrofoiling vessel shares similarities with air planes regarding their dynamics. Therefore, the marine research field on hydrofoiling overlaps with aerospace engineering. It is worth looking at the research on airplanes dynamics.

Aircraft flight dynamics often use a different state representation, in which the velocity is transformed into the flow velocity reference frame $\mathcal{F}$. The velocity vector $\mathbf{v}^{\mathcal{W}} = (u, v, w)$ is expressed as the true airspeed velocity V_a , angle of attack α and side-slip angle β [11](p. 43), [14](p. 53). For the transformation the velocity vector in the vessel's reference frame can be expressed as

$$\mathbf{v}^{\mathcal{B}} = \mathbf{q}^{\mathcal{W}} \odot \mathbf{v}^{\mathcal{W}}. \quad (5)$$

The resulting representation common in in the aerospace community is

$$\begin{pmatrix} V_a \\ \beta \\ \alpha \end{pmatrix} = \begin{pmatrix} \|\mathbf{v}^{\mathcal{B}}\| \\ \arcsin\left(\frac{v^{\mathcal{B}}}{V_a}\right) \\ \arctan\left(\frac{w^{\mathcal{B}}}{u^{\mathcal{B}}}\right) \end{pmatrix}. \quad (6)$$

The forces and moments are defined as products of coefficients and dimensions. The coefficients are non-dimensional and allow comparison of various aircraft configurations and scales. Using *dynamic pressure* q and *reference area* S the force vector is [14](p. 62)⁶

$$\mathbf{F}^{\mathcal{F}} = qS \begin{pmatrix} -C_D \\ C_Y \\ C_L \end{pmatrix}. \quad (7)$$

The roll, pitch and yaw moments are expressed using the wingspan b and chord length as reference lengths c

⁵Formal definitions of $\odot$ and $\otimes$ are listed in the appendix.

⁶In the aerospace community the notation for the force vector $\mathbf{F}$ commonly is $[X, Y, Z]$ and is $[L, M, N]$ for the moment $\boldsymbol{\tau}$.

$$\boldsymbol{\tau}^{\mathcal{F}} = qS \begin{pmatrix} C_l b \\ C_m c \\ C_n b \end{pmatrix}. \quad (8)$$

The dynamic pressure is defined as [14], [15](p. 61)

$$q = \frac{1}{2} \rho V_a^2 \quad (9)$$

where ρ is the fluid density. The aerodynamic force coefficients are dependent on the airflow angles, angular rates and control surfaces [14], [15](p. 62). For classical airplanes and small perturbations the aerodynamic force and moment terms are assumed to only be dependent on the disturbed motion and control. They are then linearized using the first order Taylor expansion of each partial derivative [16](p. 83), [15](p. 200).

This method has been applied to single track solar boats [17], [18] and sailing catamarans [19].

2.1.8.2. Hydrofoil Specialties

The airplane dynamics do not completely map the behavior of hydrofoils. A phenomenon called *ventilation* occurs, when air accumulates on the lower pressure side of a lift surface. This abruptly reduces the lift fore on the wing and occurs if a submerged foil moves close to the water surface or if a surface piercing foil or rudder experiences a very low pressure due to an angle of attack of higher than five to six degrees [11](p. 178).

A second effect that only occurs with foils in fluid is *cavitation*. If the dynamic pressure on the lifting surface is lower than the vapor pressure of the fluid, the fluid starts to boil, forming vapor-filled cavities. This reduces lift and increases drag [11](p. 169).

When a submerged foil is close to the surface, the low pressure region above the foil creates a wake, creating a free surface effect decreasing lift [11](p. 170).

Hydrofoiling vessel such as described in Section 2.1.2 and some hydrofoiling solar boats do not have fixed wings and control surfaces like a typical aircraft [17], [18]. Instead of control surfaces, the entire lifting surfaces are actuated.

The above-mentioned effect is not regarded by models described in Section 2.1.8.1. There is research on more advanced methods to model hydrofoil vessels, which addresses 3D flow, foil interaction, surface effects and wave influence [20]. This model has been applied for modeling hydrofoiling ferries [21]. For the model derivation in this thesis these spetial effect are asumend to be very small and not grantly change the dynamic behaviour.

2.1.9. Thrust Model

The model of the thrust maps the mechanical power P , input u_{thrust} to a force applied at the thruster. There are different approaches to model the hydrodynamic behavior.

2.1.9.1. Power Relation

The dynamics of a propeller in a fluid are very complex. Therefore, a rudimentary linear approach based on power can be examined. Mechanical work is defined by $W = F \cdot s$ with F being a force and s a displacement. Applying the time derivative

$$\frac{dW}{dt} = F \cdot \frac{ds}{dt} \quad (10)$$

yields the instantaneous power

$$P = F \cdot v. \quad (11)$$

Using the relationship of force and power from Equation 11, limiting the maximum static thrust by $F_{\max}$ and substituting v for the advance velocity V_a the thrust of the vessel can be expressed with

$$F_{\text{thrust}} = \min \left\{ \frac{P}{V_a}, F_{\max} \right\}. \quad (12)$$

2.1.9.2. Actuator Disk Theory

This theory is interpreting the propeller as a disc at which the fluid flows separated from the surrounding flow. Details such as the exact flow over the propeller blades are being neglected. With the advance velocity V_a , propeller cross section A , and fluid density ρ , flow continuity and momentum change lead to the thrust F_{thrust} being the roots of [14](p. 107)

$$F_{\text{thrust}}^3 + 2\rho A V_a P F_{\text{thrust}} + 2\rho A P^2 = 0. \quad (13)$$

2.1.9.3. Propeller with Coefficient

A more sophisticated approach describes the thrust

$$F_{\text{thrust}} = C_T \rho V_a^2 D^2 \quad (14)$$

where the coefficient C_T is a function of advance coefficient $J = \frac{V_a}{nD}$ with rotational speed n , Reynolds number $R_n = \frac{\rho n D^2}{\mu}$, cavitation number σ_0 [22]:

$$C_T = \varphi(J, R_n, \sigma_0). \quad (15)$$

The coefficient C_T is determined experimentally or with computational fluid dynamics methods.

2.2. Motion Planning

This section will give an overview of the methodologies used in-motion planning, focusing on the algorithms used in this thesis. Motion planning typically involves navigating through a workspace while adhering to constraints, such as initial and final positions and configurations and obstacles. This work distinguishes between trajectory planning and optimal control problems. Planning is considered a high level process, planning a trajectory while control addresses optimal following of that trajectory on a lower level.

Let us define a motion planning problem in a configuration $\mathcal{X}$ that specifies all possible configurations of the robot with constraints $\mathcal{X}_{\text{obs}} \neq \mathcal{X}_{\text{free}} \subset \mathcal{X}$ as finding a continuous path $\tau : [0, 1] \rightarrow \mathcal{X}_{\text{free}}$ such that $\tau(0) = \mathbf{x}_s$ with start configuration $\mathbf{x}_s \in \mathcal{X}_{\text{free}}$ and $\tau(1) = \mathbf{x}_g$ goal configuration $\mathbf{x}_g \in \mathcal{X}_{\text{free}}$ [23].

There are two essential criteria for planning algorithms: *completeness* and *optimality*. If a motion planner is *complete* the algorithm either finds a solution or will correctly report in finite time that no solution exists. A *probabilistically complete* planner that converges to a solution, provided that exists. *Optimality* applies if an algorithm finds the optimal solution regarding a cost function. If the solution of the planner converges to the optimal path it is *asymptotically optimal*. An *admissible* planner is complete and optimal resulting provably the best algorithm.

2.2.1. Graph Based Motion Planning

Graph Based Motion Planning discretizes the configuration space and constructs a graph to move from one discrete configuration to another. Then shortest path graph algorithms like A* are used to find the optimal trajectory subject to a heuristic function. A* is admissible.

2.2.2. Kinodynamic Motion Planning

Kinodynamic motion planning addresses the challenge of generating collision-free and physically feasible trajectories for robots operating under second-order dynamics, where acceleration constraints, forces and torques govern motion, unlike geometric planning, which neglects such temporal dependencies. This approach operates in a higher-dimensional state space (combining configuration $\mathbf{x}$, its derivative $\dot{\mathbf{x}}$) to ensure solutions that satisfy both spatial obstacle avoidance and system-level constraints, making it essential for real-world robotic systems with dynamic interactions. While graph-based methods like state lattices with motion primitives can discretize this complex space for kinodynamic planning, their computational demands escalate significantly [24].

2.2.3. Sampling Based Motion Planning

In contrast to graph-based methods, *sampling-based planners* address the challenge of high-dimensional configuration spaces by directly sampling the free configuration space $\mathcal{X}_{\text{free}}$ to construct feasible paths. While this approach offers significant speed advantages, especially for complex, high-dimensional problems, sampling-based planners are inherently *incomplete* (i.e., they may fail to find a solution even when one exists) [25]. A very popular algorithm is *Probabilistic Roadmap* (PRM). This algorithm first “learns” the configuration space, by randomly sampling the configuration space and constructing a road map of vertices as collision free configurations and edges as feasible paths between them. These paths are generated by a local planner. Then the start and goal configuration is added to the road map and a path finding algorithm searches for a solution [26]. One of the most efficient motion planning algorithms is *Rapidly-exploring Random Trees* (RRT). RRT operates by incrementally growing a tree $\mathcal{T}$

of size K from an initial configuration $\mathbf{x}_s$ through random sampling. Figure 3 shows the algorithm to build a tree with a time step resolution of Δt . Each new vertex is added only if both the vertex itself and the edge connecting it to the existing tree lie entirely within $\mathcal{X}_{\text{free}}$. This probabilistic exploration ensures robust navigation in cluttered environments but sacrifices optimality [27].

```

GENERATE_RRT( $\mathbf{x}_s, K, \Delta t$ ):
1   $\mathcal{T}.\text{init}(\mathbf{x}_{\text{init}})$ 
2  for  $k = 1$  to  $K$  do
3     $\mathbf{x}_{\text{rand}} \leftarrow \text{Random\_State}()$ 
4     $\mathbf{x}_{\text{near}} \leftarrow \text{Nearest\_Neighbour}(\mathbf{x}_{\text{rand}}, \mathcal{T})$ 
5     $\mathbf{u} \leftarrow \text{Select\_Input}(\mathbf{x}_{\text{rand}}, \mathbf{x}_{\text{near}})$ 
6     $\mathbf{x}_{\text{new}} \leftarrow \text{New\_State}(\mathbf{x}_{\text{near}}, \mathbf{u}, \Delta t)$ 
7    if  $\text{Collision\_Free}(\mathbf{x}_{\text{near}} \rightarrow \mathbf{x}_{\text{rand}})$ 
8       $\mathcal{T}.\text{add\_vertex}(\mathbf{x}_{\text{new}})$ 
9       $\mathcal{T}.\text{add\_edge}(\mathbf{x}_{\text{near}}, \mathbf{x}_{\text{new}}, \mathbf{u})$ 
10 return  $\mathcal{T}$ 

```

Figure 3: Pseudo-code for RRT generation with start configuration $\mathbf{x}_s$, K iterations and step size Δt

For kinodynamic systems—which incorporate second order dynamics—*kinodynamic RRT* extends RRT by integrating the state transition equation $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$ (described in Section 2.1.5) over a fixed time interval Δt with control inputs $\mathbf{u}$ [28].

The algorithm *Stable Sparse RRT* (SST) enables asymptotically optimal kinodynamic planning without requiring a steering function. This is achieved by the utilization of a best-first selection strategy and a pruning process. The selection parameter δ_{BN} influences the number of nodes that are considered when selecting nodes to expand. This should not be too high as it could prevent the algorithm from discovering narrow passages. δ_s controls the tuning process. The larger the value, more states are pruned. A too large value will result in poor exploration of the state space [29]. Figure 4 depicts a high level overview of how the algorithm works. The `Best_First_Selection_SST` function selects the state with the best path instead of the nearest to the sampled configuration. Then `MonteCarlo-Prop` generates new state by random sampling of actions. The `CollisionFree` function checks if the propagation is collision free. `Is_Node_Locally_the_Best_SST` checks if the new state is the best in the current graph inside a given radius. If that is the case, other states within the radius are pruned with the `Prune_dominant_Nodes_SST` function. The *SST** is an extension to SST, that automatically tunes these hyperparameters, decreasing them over time.

```

STABLE_SPARSE_RRT( $\mathcal{X}, \mathcal{U}, \mathbf{x}_s, \Delta t, N, \delta_{\text{BN}}, \delta_s$ ):
1   $\mathbb{V}_{\text{active}} \leftarrow \{\mathbf{x}_s\}, \mathbb{V}_{\text{inactive}} \leftarrow \emptyset;$ 
2   $G = \{V \leftarrow (\mathbb{V}_{\text{active}} \cup \mathbb{V}_{\text{inactive}}), \mathbb{E} \leftarrow \emptyset\};$ 
3   $s_0 \leftarrow x_0, s_0.\text{rep} = \mathbf{x}_s, S \leftarrow \{s_0\}$ 
4  for  $N$  iterations do
5     $x_{\text{selected}} \leftarrow \text{Best\_First\_Selection\_SST}(\mathcal{X}, \mathbb{V}_{\text{active}}, \delta_{\text{BN}});$ 
6     $x_{\text{new}} \leftarrow \text{MonteCarlo\_Prop}(x_{\text{selected}}, \mathcal{U}, \Delta t);$ 
7    if  $\text{CollisionFree}(x_{\text{selected}} \rightarrow x_{\text{new}})$  then
8      if  $\text{Is\_Node\_Locally\_the\_Best\_SST}(x_{\text{new}}, S, \delta_s)$  then
9         $\mathbb{V}_{\text{active}} \leftarrow \mathbb{V}_{\text{active}} \cup \{x_{\text{new}}\};$ 
10        $\mathbb{E} \leftarrow \mathbb{E} \cup (x_{\text{selected}} \rightarrow x_{\text{new}});$ 
11        $\text{Prune\_dominated\_Nodes\_SST}(x_{\text{new}}, \mathbb{V}_{\text{active}}, \mathbb{V}_{\text{inactive}}, \mathbb{E});$ 
return  $G;$ 

```

Figure 4: High level pseudo-code of the SST Algorithm with configuration space $\mathcal{X}$, action space $\mathcal{U}$, step size Δt , N iterations, and mentioned selection parameter δ_{BN} and pruning parameter δ_s

2.2.4. Optimization Based Planning

Optimization Based Methods formulate the motion planning problem as an optimization problem of a cost function J . When it may not be feasible to compute the gradient of the optimization problem *gradient-free* methods like *Shortcutting* can be applied. *Gradient-based* methods use the gradient ∇J to improve the path by deforming it into the direction of lower cost. *Covariant Hamilton Optimization for Motion Planning* (CHOMP) discretizes the path into waypoints and iterative updates them based on the gradient ∇J converging to a local minimum [30].

2.2.5. Applications

Motion planning has been applied to a variety of different robots. There has been research on motion planning for a fixed wing unmanned aerial vehicle (UAV) with a twelve-dimensional state space and comparable dynamics. They used SST for planning a collision free route avoiding obstacles and respecting the flight envelope [5].

In the maritime sector motion planning is used to find routes optimizing for different criteria such as fuel consumption or duration. There is ongoing research on path planning in polar ice regions [31].

2.3. Control

This section will first give a short introduction into control theory. As a next step, focus on the methods used in this thesis and give an overview of related work regarding control and hydrofoiling vessels.

Control theory provides the mathematical framework for designing systems to maintain desired behavior influenced by uncertainties, disturbances, and dynamic constraints. At its core, it addresses the question, how to manipulate system inputs $\mathbf{u}$ to achieve target outputs $\mathbf{y}$ while ensuring *stability* and *robustness*. In control engineering a system is called stable if the dynamics respond with a bounded output to a bounded input [32](p. 395). A robust system is maintaining an acceptable stability performance under disturbance, noise and model uncertainties [32](p. 883).

Classical control theory analyses systems in *frequency domain* consisting of magnitude and phase. Control methods such as root locus, Bode or Nyquist use output feedback [32](p. 447,552,623). These methods have limitations: They cannot directly influence the internal dynamics due to not having information about the full system state. They are impractical for multiple input multiple output systems and lack a systematic way to handle state and input constraints.

The time-domain approach uses a state-space representation where the system's state of Order n is represented by a time-dependent state vector $\mathbf{x}(t) \in \mathbb{R}^n$, whose elements completely describe a system at a given point in time t . The first order non-linear differential equation, called dynamic function

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t)), \quad \mathbf{x}(t_0) = \mathbf{x}_0 \quad (16)$$

describes the change rate of the states dependent of the state itself and the input vector $\mathbf{u}(t)$. The non-linear measurement function

$$\mathbf{y}(t) = \mathbf{h}(\mathbf{x}(t), \mathbf{u}(t)) \quad (17)$$

describes how the measurements $\mathbf{y}(t)$ depend on the state and input. Assuming that parameters of the systems are time invariant, and the differential equation can be approximately linearized, the system can be written as

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \\ \mathbf{y}(t) &= \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t), \end{aligned} \quad (18)$$

with state matrix $\mathbf{A}$, input matrix $\mathbf{B}$, output matrix $\mathbf{C}$ and feed through $\mathbf{D}$. These system are also known as linear time-invariant Systems (LTI-Systems).

2.3.1. Observer

As direct measurement is often not possible in real-world models, observers are employed to estimate the unmeasured states from available inputs $\mathbf{u}$ and outputs $\mathbf{y}$. An observer fuses the plant model with the measured data to produce estimated systems state vector $\hat{\mathbf{x}}$. These estimated states enable advanced control methods such as *linear quadratic regulator* (LQR) or *model predictive control* (MPC). Common approaches for state estimation is the Kalman Filter and its extended and unscented expansions and Moving Horizon Estimation [33], [34].

2.3.2. Controllability

It is of great interest to ensure that all states of a system are controllable. The definition of *Controllability* is as follows: if there is exist an unconstrained input trajectory that can transfer any initial state into any other state in finite time, the system is *controllable*. For LTI-Systems this is equivalent to the expression [32]

$$\text{rank}[A \quad AB \quad A^2B \dots A^{n-1}B] = n. \quad (19)$$

2.3.3. Optimal Control

Advanced controller design methods frame the problem of finding policy $\pi(\mathbf{x}) = \mathbf{u}^*$ as an optimization problem [35](p. 220). We seek to find the policy that minimizes some objective function J :

$$\pi = \text{argmin } J(\pi). \quad (20)$$

2.3.4. Linear Quadratic Regulator

The objective function J typically encodes a cost for tracking performance, energy consumption, stability margins, robustness to disturbances, and satisfaction of constraints. A common form of an objective function is the quadratic cost function

$$J = \frac{1}{2} \int_0^\infty [\mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{u}^\top \mathbf{R} \mathbf{u}] dt \quad \min_{\mathbf{u}} J(\mathbf{u}) \Rightarrow \mathbf{u}^* \quad (21)$$

where $\mathbf{Q}$ and $\mathbf{R}$ are weighting matrices that allow to tradeoff between state regulation and control effort [11](p. 419),[35](p. 132).

For observable LTI systems the optimal control problem can be solved with the variational method. This method yields $\mathbf{P}$ as the positive definite solution to the algebraic Riccati differential equation

$$\mathbf{A}^\top \mathbf{P} + \mathbf{P} \mathbf{A} - \mathbf{P} \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^\top \mathbf{P} + \mathbf{Q} = 0. \quad (22)$$

The optimal input $\mathbf{u}^*$ is defined as [36](p .113)

$$\mathbf{u}^* = -\mathbf{K} \cdot \mathbf{x} = -\mathbf{R}^{-1} \mathbf{B}^\top \mathbf{P} \cdot \mathbf{x}. \quad (23)$$

This method is well-known as the *linear quadratic regulator* (LQR). In application the main challenge is tuning the $\mathbf{Q}$ and $\mathbf{R}$ matrices. Large entries in $\mathbf{Q}$ penalize large value in the state vector $\mathbf{x}$. This results in fast decrease of state values at the cost of large inputs. The matrix $\mathbf{R}$ penalizes large inputs and therefore decreases the states rate of change. Usually only the diagonal elements of $\mathbf{Q}$ and $\mathbf{R}$ are nonzero. Entries in the off-diagonal penalize the combination of states. The Riccati differential equation has analytical solution, so numerical methods are used for solving [36](p .115).

2.3.5. Model Predictive Control

While the linear quadratic regulator provides an optimal feedback law for linear systems with quadratic cost functions, it operates under the assumption of un-

constrained inputs and states. In many practical control problems, physical and safety constraints must be enforced. *Model predictive control* (MPC) addresses this limitation by formulating the control task as a finite-horizon optimal control problem that is solved repeatedly in real time. At each sampling instant, MPC predicts the future of the system states over a finite horizon using a dynamic model of the system. Based on this prediction, it computes an optimal sequence of control inputs $\mathbf{u}^*$ that minimize a predefined cost function, with similar structure of Equation 21. Only the first input of this optimal sequence is applied to the system, and the procedure is repeated at the next sampling step in a receding-horizon fashion [37](p. 1f). A key advantage of MPC lies in its ability to explicitly handle input constraints

$$\|\mathbf{u}_i\| \leq \mathbf{u}_{\max,i} \quad \forall p \in \{0, \dots, p\} \quad (24)$$

and state constraints

$$\|\mathbf{x}_i\| \leq \mathbf{x}_{\max,i} \quad \forall q \in \{0, \dots, q\} \quad (25)$$

while maintaining good tracking performance [37](p. 6f). Regular MPC is optimizing with regard to a LTI-System of the form defined in Equation 18. In order to deploy linear MPC for non-linear Systems the dynamics are linearized at the reference. Another approach is *nonlinear model predictive control* NMPC. NMPC can handle non-linear dynamics without linear approximations [38].

2.3.6. Related Work on Control for Hydrofoil Vessels

General control of hydrofoiling vessels is a well-studied field. A great variety of control methods have been applied. Classical methods like pole placement give good results for control a single track solar hydrofoil vessel [39].

In recent years, research on machine learning methods for hydrofoil control has increased. Reinforcement learning is applied to control for solar boats with mixed results [40]. Moreover, it is researched for active wave control for hydrofoil ferries with an advanced dynamic model [41]. The application of a radial basis function neural network is explored for ferries as well [42]. Other data-driven approaches like Jacobian learning are deployed for full state control of the Americas Cup 75 hydrofoiling sailing catamarans. For Americas Cup 72 vessels optimal control methods such as LQR have been applied [19].

The research mentioned above focusses on stability at an operating point. Trajectory following control for hydrofoiling vessels is a significantly less researched subject. There are publications for path-following control of surface vessels using MPC in three degrees of freedom [43], [44]. Studies of path-following control in six degrees of freedom using MPC can only be found for airplanes [5].

3. Approach

3.1. Modeling the Dynamics

3.1.1. Approach for Hydro Dynamic Forces

The forces of foils, masts and the rudder define the dynamic behavior of the vessel. The force at each hydrodynamic surface is mostly depending on angle of attack and velocity. The complex dynamics arise from the different velocity vectors at the foil locations. The angular rates induce tangential velocities, that increase or decrease the angle of attack depending on the location. Models described in Section 2.1.8.2 derive their motion equations by using constant coefficients. This creates a model that has linear forces and moments with respect to the second order states and inputs as well as state couplings. Each dynamic effect is linearized with a corresponding constant hydrodynamic coefficient. This results in specific lift coefficient, angle of attack coefficients, roll pitch yaw coefficients. These coefficients must be determined experimentally with system identification or with elaborate simulation methods.

By contrast, in this thesis a different approach is developed. The mentioned linear models have been developed for control of new steady-state analysis. This is not sufficient for motion planing with harsh maneuvers. By transforming each foil into its flow reference frame, calculating the forces using thin airfoil theory and transform back, a more accurate non-linear model can be derived. This can be achieved by modeling stalling of foils at high angle of attack in extreme maneuvers and that without the need for the specific coefficients which are cumbersome to obtain. The linearization for linear controllers can then be derived from a computer algebra system. An additional advantage is that any foil configuration can be simulated with this method.

3.1.2. Hydrodynamic Force and Moment

The three lifting foils, rudder and the two front masts contribute to the total force and torque on the vessel. Each hydrodynamic surface i has a position $\mathbf{p}_i^{\mathcal{B}}$ relative to the center of gravity an orientation $\mathbf{q}_i^{\mathcal{B}}$ relative to the fixed body coordinates, chord length c_i , reference area S_i and may be actuated in pitch by the angle γ_i . The resulting force and moment of the foil is also dependent of the state $[\mathbf{p}^{\mathcal{W}}, \mathbf{q}^{\mathcal{W}}, \mathbf{v}^{\mathcal{W}}, \boldsymbol{\omega}^{\mathcal{B}}] = \mathbf{x} \in X$.

For application of the aerodynamic theory in Section 2.1.8.1, the fluid velocity must be transformed into the flow reference frame. The global velocity of the foil is the vessel's velocity $\mathbf{v}^{\mathcal{W}}$ with addition of tangential velocity due to the angular velocity of the vessel:

$$\mathbf{v}_i^{\mathcal{W}} = \mathbf{v}^{\mathcal{W}} + (\mathbf{q}^{\mathcal{W}})^{-1} \odot (\boldsymbol{\omega}^{\mathcal{B}} \times \mathbf{p}_i^{\mathcal{B}}). \quad (26)$$

Applying the vessels rotation and the foil rotation yields the velocity

$$\mathbf{v}_i^{\mathcal{F}_i} = \mathbf{q}_i^{\mathcal{B}} \otimes \mathbf{q}^{\mathcal{W}} \odot \mathbf{v}_i^{\mathcal{W}} \quad (27)$$

in the fixed foil reference frame $\mathcal{F}_i$.

In similar fashion to Equation 6 the velocities can be expressed in the flow coordinates total velocity V_i , sideslip angle β_i , angle of attack α_i :

$$\begin{pmatrix} V_i \\ \beta_i \\ \alpha_i \end{pmatrix} = \begin{pmatrix} \|\mathbf{v}_i^{\mathcal{F}_i}\| \\ \arcsin\left(\frac{v_i^{\mathcal{B}}}{V_i}\right) \\ \arctan\left(\frac{w_i^{\mathcal{B}}}{u_i^{\mathcal{B}}}\right) \end{pmatrix}. \quad (28)$$

The lift F_L and drag F_D are calculated with their aerodynamic coefficient functions, offsetting the angle of attack by the actuated pitch angle γ_i ,

$$F_L = C_{L_i}(\alpha_i + \gamma_i)q_i S_i \quad F_D = C_{D_i}(\alpha_i + \gamma_i)q_i S_i \quad (29)$$

with q_i as the dynamic pressure

$$q_i = \frac{1}{2}\rho V_i^2. \quad (30)$$

As the foils chord runs parallel to the y-plane, there are no lateral forces. The force vector can be written as

$$\mathbf{F}_i^{\mathcal{F}_i}(\alpha_i, V_i, S_i) = \mathbf{R}_y(\alpha_i) \frac{1}{2}\rho V_i^2 S_i \begin{pmatrix} -C_D(\alpha_i + \gamma_i) \\ 0 \\ C_L(\alpha_i + \gamma_i) \end{pmatrix}, \quad (31)$$

where

$$\mathbf{R}_y(\alpha) = \begin{pmatrix} \cos \alpha & 0 & \sin \alpha \\ 0 & 1 & 0 \\ -\sin \alpha & 0 & \cos \alpha \end{pmatrix} \quad (32)$$

is the rotation matrix round the y-axis, transforming the lift and drag forces from the flow reference frame into the foils reference frame $\mathcal{F}_i$. The drag force is acting in the opposite direction of flow, hence the negative sign.

For application of the forces onto the rigid body dynamics the force is transformed back into the forces reference frame. The final force of a foil can be written as a function depending on the state x and the foil parameters

$$\mathbf{F}_i^{\mathcal{B}}(\mathbf{x}, \mathbf{p}_i, \mathbf{q}_i, c_i, S_i, \gamma_i) = \mathbf{q}_i^{-1} \odot \mathbf{F}_i^{\mathcal{F}_i}. \quad (33)$$

The intentions torque on the rigid body generated by the force is defined with the cross product of the position of the foil as the lever around the center of

mass:⁷

$$\tau_i^{\mathcal{B}}(\mathbf{x}, \mathbf{p}_i, \mathbf{q}_i, c_i, S_i, \gamma_i) = \mathbf{q}_i^{-1} \odot (\mathbf{p}_i \times (\mathbf{F}^{\mathcal{F}_i})). \quad (34)$$

The total force and total torque are the sum of all the six foils depicted in Figure 2 acting on the vessel. Each foil has a set of parameters and input actuation. The front masts are not actuated and therefor have a constant pitch angle of zero:

$$\begin{aligned} \mathbf{F}_{\text{foil}} = & \mathbf{F}_{\text{stb, lift}}^{\mathcal{B}}(\mathbf{x}, \mathbf{p}_{\text{stb, lift}}, \mathbf{q}_{\text{stb, lift}}, c_{\text{stb, foil}}, S_{\text{stb, foil}}, u_{\text{stb}}) \\ & + \mathbf{F}_{\text{port, lift}}^{\mathcal{B}}(\mathbf{x}, \mathbf{p}_{\text{port, lift}}, \mathbf{q}_{\text{port, lift}}, c_{\text{port, lift}}, S_{\text{port, lift}}, u_{\text{port}}) \\ & + \mathbf{F}_{\text{stern, lift}}^{\mathcal{B}}(\mathbf{x}, \mathbf{p}_{\text{stern, lift}}, \mathbf{q}_{\text{stern, lift}}, c_{\text{stern, lift}}, S_{\text{stern, lift}}, u_{\text{stern}}) \\ & + \mathbf{F}_{\text{rudd}}^{\mathcal{B}}(\mathbf{x}, \mathbf{p}_{\text{rudd}}, \mathbf{q}_{\text{rudd}}, u_{\text{rudd}}, S_{\text{rudd}}, u_{\text{rudd}}) \\ & + \mathbf{F}_{\text{stb, mast}}^{\mathcal{B}}(\mathbf{x}, \mathbf{p}_{\text{stb, mast}}, \mathbf{q}_{\text{stb, mast}}, u_{\text{stb, mast}}, S_{\text{stb, mast}}, 0) \\ & + \mathbf{F}_{\text{port, mast}}^{\mathcal{B}}(\mathbf{x}, \mathbf{p}_{\text{port, mast}}, \mathbf{q}_{\text{port, mast}}, u_{\text{port, mast}}, S_{\text{port, mast}}, 0) \end{aligned} \quad (35)$$

$$\begin{aligned} \tau_{\text{foil}} = & \tau_{\text{stb, lift}}^{\mathcal{B}}(\mathbf{x}, \mathbf{p}_{\text{stb, lift}}, \mathbf{q}_{\text{stb, lift}}, u_{\text{stb, lift}}, S_{\text{stb, lift}}, u_{\text{stb}}) \\ & + \tau_{\text{port, lift}}^{\mathcal{B}}(\mathbf{x}, \mathbf{p}_{\text{port, lift}}, \mathbf{q}_{\text{port, lift}}, u_{\text{port, lift}}, S_{\text{port, lift}}, u_{\text{port}}) \\ & + \tau_{\text{stern, lift}}^{\mathcal{B}}(\mathbf{x}, \mathbf{p}_{\text{stern, lift}}, \mathbf{q}_{\text{stern, lift}}, u_{\text{stern, lift}}, S_{\text{stern, lift}}, u_{\text{stern}}) \\ & + \tau_{\text{rudd}}^{\mathcal{B}}(\mathbf{x}, \mathbf{p}_{\text{rudd}}, \mathbf{q}_{\text{rudd}}, u_{\text{rudd}}, S_{\text{rudd}}, u_{\text{rudd}}, S_{\text{rudd}}, u_{\text{rudd}}) \\ & + \tau_{\text{stb, mast}}^{\mathcal{B}}(\mathbf{x}, \mathbf{p}_{\text{stb, mast}}, \mathbf{q}_{\text{stb, mast}}, u_{\text{stb, mast}}, S_{\text{stb, mast}}, 0) \\ & + \tau_{\text{port, mast}}^{\mathcal{B}}(\mathbf{x}, \mathbf{p}_{\text{port, mast}}, \mathbf{q}_{\text{port, mast}}, u_{\text{port, mast}}, S_{\text{port, mast}}, 0) \end{aligned} \quad (36)$$

3.1.3. Thrust Force and Moment

For the thrust model Equation 12 is chosen, and the advance velocity is assumed as $V_a = \|\mathbf{v}\|$. The thrust force vector in fixed body reference is defined as

$$\mathbf{F}_{\text{thrust}} = \left(\cos\left(\frac{u_{\text{rudd}}}{2}\right), 0, 0, \sin\left(\frac{u_{\text{rudd}}}{2}\right) \right)_{\mathbb{H}} \otimes e_x \min\left\{ \frac{u_{\text{thrust}}}{\|\mathbf{v}\|}, F_{\text{max}} \right\} \quad (37)$$

using a quaternion to model the thrust vectoring due to rudder input.

The moment due to thrust is derived with the thruster position $\mathbf{p}_{\text{thrust}}$ as a lever:

$$\tau_{\text{thrust}} = \mathbf{p}_{\text{thrust}} \times \mathbf{F}_{\text{thrust}} \quad (38)$$

The sum of the hydrodynamic force and thrust force

$$\mathbf{F} = \mathbf{F}_{\text{foil}}, \mathbf{F}_{\text{thrust}} \quad (39)$$

and the sum of the moments

$$\tau = \tau_{\text{foil}}, \tau_{\text{thrust}} \quad (40)$$

⁷The hydrodynamic moment acting on the foil itself due to coefficient C_m are very small in comparison to moments created by the lift forces applied by levers. Therefore they are ignored here.

complete the definition of the dynamics. All constant parameters are later defined in Table 3 in the Appendix.

3.1.4. Integration

For integration of the dynamics the explicit Euler method is chosen. One time step Δt is defined as

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k) \cdot \Delta t. \quad (41)$$

To ensure the properties of the unit quaternion, the quaternion is renormalized at every time step.

3.2. Problem Formulation

This chapter deals with the specific definition of the problems to be solved. There is a motion planning problem to find a good path and a control problem to follow the planned trajectory.

3.2.1. Motion Planning

The vessels configuration or state $\mathbf{x}$ is minimally described inside the configuration space $\mathcal{X}$. The configuration space is a compound space of position $\mathbf{p} = [x, y, z] \in \mathbb{R}^3$, orientation $\mathbf{q} = [\varphi, \theta, \psi] \in \mathbb{R}^3$ as Euler angles⁸, velocity $\mathbf{v} = [u, v, w] \in \mathbb{R}^3$ and angular velocity $\boldsymbol{\omega} = [\omega_x, \omega_y, \omega_z] \in \mathbb{R}^3$. Therefore, the configuration space is $\mathcal{X} = \mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}^3$. To ensure that the vessel stays on foil, there are some hard limits to the configuration space. The ride height is limited by z_{low} and z_{high} . The attitude is constrained in roll φ_{max} and pitch θ_{max} . The starting configuration is $\mathbf{x}_{\text{start}} \in \mathcal{X}$. The goal region

$$\mathcal{X}_{\text{goal}} = \left\{ \mathbf{x} \in \mathcal{X} : (\mathbf{x} - \mathbf{x}_{\text{goal}})^\top \mathbf{W} (\mathbf{x} - \mathbf{x}_{\text{goal}}) \leq \mu^2 \right\} \subset \mathcal{X} \quad (42)$$

is a hyper-ellipsoid around goal state $\mathbf{x}_{\text{goal}}$ with size μ and state weight matrix $\mathbf{W} \in \mathbb{R}^{12 \times 12}$. The vessel operates in the 3D Euclidean workspace $\mathcal{W} = \mathbb{R}^3$. The workspace is obstructed by n spherical obstacles $S = (\mathbf{p}, r)$

$$\mathcal{O}_{S_i} = \left\{ \mathbf{p} \in \mathcal{W} : \|\mathbf{p} - \mathbf{p}_{S_i}\| \leq r_{S_i} \right\} \quad (43)$$

and m axis aligned bounding boxes define by a minimum and maximum corner point $B = (\mathbf{a}, \mathbf{b})$

$$\mathcal{O}_{B_i} = \left\{ \mathbf{p} \in \mathcal{W} : \mathbf{p} > \mathbf{a}_{B_i}, \mathbf{p} < \mathbf{b}_{B_i} \right\}. \quad (44)$$

The obstacle free workspace

$$\mathcal{W}_{\text{free}} = \mathcal{W} \setminus \left(\mathcal{O}_{S_1} \cup \mathcal{O}_{S_2} \dots \mathcal{O}_{S_n} \right) \setminus \left(\mathcal{O}_{B_1} \cup \mathcal{O}_{B_2} \dots \mathcal{O}_{B_m} \right) \quad (45)$$

is written as the workspace without the obstacles. The configuration map

⁸The motion planning is done in Euler angles. The pitch and roll angles are small, thus preventing gimbal lock.

$$\mathcal{B} : \mathcal{X} \rightarrow \mathcal{W} \quad B(x) = \{p \in \mathbb{R}^3 : \|p - x_p\| < r\} \quad (46)$$

defines the vessel as a sphere of radius r in the workspace based on the position in configuration space. The dynamics function $\mathbf{f} : \mathcal{U} \times \mathcal{X} \rightarrow \mathcal{X}$ described in Section 2.1 maps a pair of a state $x \in \mathcal{X}$ and controls $u \in \mathcal{U}$ to its derivative. The planned trajectory is a function of controls $u : [0, T) \rightarrow \mathcal{U}$ and configurations $q : [0, T] \rightarrow \mathcal{Q}$ over the time duration T .

As a metric to evaluate the quality of the trajectory, the cost function

$$J : \mathcal{U} \times \mathcal{X} \times \mathbb{R} \rightarrow \mathbb{R} \quad (47)$$

is defined. With state cost $\mathbf{Q}$, control cost $\mathbf{R}$ and time cost s the cost function is defined as:

$$J(x(t), u(t), T) = \int_0^T x(t)^\top \mathbf{Q} x(t) + u(t)^\top \mathbf{R} u(t) dt + sT. \quad (48)$$

The motion planning problem can be described as an optimization problem, minimizing the cost with respect to start, goal, obstacles, and feasible dynamics:

$$\begin{aligned} \underset{T, u(t), x(t)}{\operatorname{argmin}} \quad & J(T, u(t), x(t)) \quad \text{s.t.} \quad x(0) = x_{\text{start}} \\ & x(T) \in \mathcal{X}_{\text{goal}} \\ & B(x(t)) \subset \mathcal{W}_{\text{free}} \quad \forall t \in (0, T) \\ & \dot{x} = \mathbf{f}(x(t), u(t)) \quad \forall t \in [0, T) \\ & z_{\text{low}} \leq x_z(t) \leq z_{\text{high}} \quad \forall t \in [0, T) \\ & |x_\varphi(t)| \leq \varphi_{\text{max}} \quad \forall t \in [0, T) \\ & |x_\theta(t)| \leq \theta_{\text{max}} \quad \forall t \in [0, T) \\ & u(t) \in \mathcal{U} \quad \forall t \in [0, T). \end{aligned} \quad (49)$$

The planning algorithms used in this thesis are defined with discrete time. Therefore, the optimization problem must be defined in a discrete time domain. The trajectories are discretized with l steps with a time resolution of Δt .

The time discrete action u_k and state x_k are samples of the continuous functions as time step k .

The discrete function $\tilde{\mathbf{f}}$ propagates the dynamics by Δt with the given state x_k and inputs u_k

$$x_{k+1} = \tilde{\mathbf{f}}(x_k, u_k). \quad (50)$$

The integral of the time continuous cost function is replaced with sum resulting to

$$\bar{J} = \sum_{k=0}^l x_k^\top \mathbf{Q} x_k + \sum_{k=0}^{l-1} u_k^\top \mathbf{R} u_k + s \cdot l \cdot \Delta t. \quad (51)$$

Based on the definitions and assumptions, the discrete optimization problem can be formulated as follows:

$$\begin{aligned}
 \underset{T, \mathbf{u}_{0:N-1}, \mathbf{x}_{0:N}}{\operatorname{argmin}} \quad & \bar{J} \quad \text{s.t.} \quad \mathbf{x}_0 = \mathbf{x}_{\text{start}}, \quad \mathbf{x}_l \in \mathbf{x}_{\text{goal}} \\
 & \mathbf{x}_{k+1} = \tilde{\mathbf{f}}(\mathbf{u}_k, \mathbf{x}_k) \quad \forall k \in \{0, \dots, l-1\} \\
 & \mathcal{B}(\mathbf{x}_k) \subset \mathcal{W}_{\text{free}} \quad \forall k \in \{0, \dots, l\} \\
 & z_{\text{low}} \leq \mathbf{x}_{k,z} \leq z_{\text{high}} \quad \forall k \in \{0, \dots, l\} \\
 & |\mathbf{x}_{k,\varphi}| \leq \varphi_{\text{max}}, \quad \forall k \in \{0, \dots, l\} \\
 & |\mathbf{x}_{k,\theta}| \leq \theta_{\text{max}} \quad \forall k \in \{0, \dots, l\} \\
 & \mathbf{u}_k \in \mathcal{U} \quad \forall k \in \{0 \dots l-1\}
 \end{aligned} \tag{52}$$

3.2.2. Control

With an optimally planned trajectory, the goal of the control problem is to find a strategy to follow the planned state reference trajectory $\mathbf{x}_{\text{ref}} : [0, T] \rightarrow \mathcal{X}$ and $\mathbf{u}_{\text{ref}} : [0, T] \rightarrow \mathcal{U}$ as best as possible, while compensating for initial state disturbances, measurement noise $\mathbf{v}(t)$ and process noise $\mathbf{w}(t)$

$$\begin{aligned}
 \dot{\mathbf{x}}(t) &= \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t)) + \mathbf{w}(t) \\
 \mathbf{y}(t) &= \mathbf{h}(\mathbf{x}(t), \mathbf{u}(t)) + \mathbf{v}(t).
 \end{aligned} \tag{53}$$

To evaluate the performance of the controller, the control cost function J_c is defined. The state error function $\mathbf{e}(t) : [0, T] \rightarrow \mathcal{E}$ describes the error between the desired state and reference state. The final cost function is

$$\begin{aligned}
 J_c &= \int_0^T \mathbf{e}(t)^\top \mathbf{Q}_c \mathbf{e}(t) \\
 &\quad + (\mathbf{u}(t) - \mathbf{u}_{\text{ref}}(t))^\top \mathbf{R}_c (\mathbf{u}(t) - \mathbf{u}_{\text{ref}}(t)) \, dt \\
 &\quad + \mathbf{e}(T)^\top \mathbf{Q}_{T,c} \mathbf{e}(T)
 \end{aligned} \tag{54}$$

with $\mathbf{Q}_c \in \mathbb{R}^{12 \times 12}$ as a state weight matrix, $\mathbf{Q}_{T,c} \in \mathbb{R}^{12 \times 12}$ as the terminal state matrix and control weights $\mathbf{R}_c \in \mathbb{R}^{5 \times 5}$. The control problem can now be defined as finding the optimal inputs trajectory that minimizes the cost function, while propagating dynamic function, and retaining input constraints:

$$\begin{aligned}
 \underset{\mathbf{u}}{\operatorname{argmin}} \quad & J_c(\mathbf{e}(t), \mathbf{u}(t), \mathbf{u}_{\text{ref}}(t)) \quad \text{s.t.} \quad \mathbf{u}(t) \in \mathcal{U} \quad \forall t \in [0, T] \\
 & \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t) + \mathbf{w}(t)) \quad \forall t \in [0, T].
 \end{aligned} \tag{55}$$

For implementation the time for the controller is discretized into time steps. The continuous references $\mathbf{x}_{\text{ref}}(t)$ and $\mathbf{u}_{\text{ref}}(t)$ are sampled with the discrete time Δt_c denoted as $\mathbf{x}_{\text{ref},k} \quad \forall k \in \{0, \dots, l\}$ and $\mathbf{u}_{\text{ref},k} \quad \forall k \in \{0, \dots, l-1\}$.

The cost function sums the weighed squared discrete state errors between the reference and actual states:

$$J_c = \sum_{k=0}^l e_k^\top Q_c e_k + \sum_{k=0}^{l-1} (\mathbf{u}_k - \mathbf{u}_{\text{ref},k})^\top \mathbf{R}_c (\mathbf{u}_k - \mathbf{u}_{\text{ref},k}). \quad (56)$$

With these adjustments the optimization problem can be reformulated for the discrete case as

$$\begin{aligned} \underset{\mathbf{u}_{0:l-1}}{\text{argmin}} J_c \quad \text{s.t.} \quad & \mathbf{u}_k \in \mathcal{U} \quad \forall k \in \{0, \dots, l\} \\ & \mathbf{x}_{k+1} = \tilde{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{w}_k \quad \forall k \in \{0, \dots, l\}. \end{aligned} \quad (57)$$

3.3. Motion Planning Methods

3.3.1. Configuration Validity

The sampling based planners described in Section 2.2.3 need a way to determine if a sampled configuration is valid. This valid set is denoted as $\mathcal{X}_{\text{valid}} \subset \mathcal{X}$. The valid configuration set is constrained by the workspace bounds, attitude constraints and obstacles.

The workspace is constrained to an axis aligned box

$$\mathcal{W} = [x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}] \times [z_{\min}, z_{\max}] \subset \mathbb{R}^3, \quad (58)$$

where x and y define the working area and $z_{\min}$ and $z_{\max}$ the safe height where the hulls do not touch the water surface and the foils do not ventilate.

The maximum roll and pitch angles prevent the foils from ventilation by piercing the water surface allowing only

$$\mathcal{X}_{\varphi, \theta} = \{\mathbf{x} \in \mathcal{X} \mid |\mathbf{q}_\varphi| < \varphi_{\max}, |\mathbf{q}_\theta| < \theta_{\max}\} \quad (59)$$

as configurations.

There are obstacles of two types: Spheres of position a radius ($\mathbf{p}_s \in \mathbb{R}^3, r_s \in \mathbb{R}$) and axis aligned bounding boxes defined by a minimum and maximum point ($\mathbf{p}_{\min} \in \mathbb{R}^3, \mathbf{p}_{\max} \in \mathbb{R}^3$). The valid states with respect to collisions are defined as

$$\begin{aligned} \overline{\mathcal{X}_{\text{collision}}} = \{[\mathbf{p}, \mathbf{q}, \mathbf{v}, \boldsymbol{\omega}] \in \mathcal{X} \mid \\ \|\mathbf{p} - \mathbf{p}_s\| > r + r_s, \quad \|\max(\mathbf{p}_{\min}, \min(\mathbf{p}, \mathbf{p}_{\max}))\| > r\}. \end{aligned} \quad (60)$$

Finally, all valid configurations exist in the intersection of the three constraints:

$$\mathcal{X}_{\text{valid}} = \{\mathbf{x} \in \mathcal{X} \mid \mathbf{p} \in \mathcal{W}\} \cap \mathcal{X}_{\varphi, \theta} \cap \overline{\mathcal{X}_{\text{collision}}}. \quad (61)$$

3.3.2. Distance Metric

The algorithms need a distance metric $d : \mathcal{X}_e \times \mathcal{X}_e \rightarrow \mathbb{R}$ for nearest neighbour calculation. Due to the fact that the configuration space containing rotation is not Euclidean, a special distance metric is to be defined. The position, velocity and angular velocity components of $\mathcal{X}$ exit in $\mathbb{R}^3$ such that the Euclidean distance norm $\|\cdot\|_2$ can be applied. As described in the previous section, the orientation is limited to small angles in roll φ and pitch θ . The distance can be

approximated by the Euclidean norm. The vessel is able to move into any yaw orientation $\psi \in [0, 2\pi)$ making it live in $\text{SO}(2)$. For this topology the geodesic distance is applied. The total distance is the sum of each component weighted by $w_p, w_a, w_\psi, w_v, w_\omega$ allowing for finetuning of distance mapping:

$$\begin{aligned} d(\mathbf{x}, \mathbf{x}')^2 = & w_p \|\mathbf{p} - \mathbf{p}'\|_2^2 \\ & + w_a (x_\varphi - x'_\varphi)^2 + w_a (x_\theta - x'_\theta)^2 \\ & + w_\psi \min\{\|x_\psi - x'_\psi\|, 2\pi - \|x_\psi - x'_\psi\|\}^2 \\ & + w_v \|\mathbf{v} - \mathbf{v}'\|_2^2 \\ & + w_\omega \|\boldsymbol{\omega} - \boldsymbol{\omega}'\|_2^2. \end{aligned} \quad (62)$$

3.3.3. Algorithms

Since it is very difficult if not impossible to derive a steering function for the dynamics, only sampling based planners can be considered to find an initial solution.

3.4. Control Methods

3.4.1. Error Function

Before solving the control problem with two different approaches, the state error function $e_k \in \mathcal{E}$ used in Equation 56 must be defined. The state is $x \in \mathcal{X}$. While the position, velocities and angular velocities are in Euclidean $\mathbb{R}^3$ the orientation mapped in non-Euclidian $\text{SO}(3)$ space. A simple subtraction of the components is not a viable metric.

Let us define the difference equivalent operator $\ominus : \mathcal{X} \times \mathcal{X} \rightarrow \mathcal{X}$ such that

$$x \ominus x' = \begin{pmatrix} \mathbf{p} - \mathbf{p}' \\ \mathbf{q}'^{-1} \mathbf{q} \\ \mathbf{v} - \mathbf{v}' \\ \boldsymbol{\omega} - \boldsymbol{\omega}' \end{pmatrix}. \quad (63)$$

The individual components of quaternion cannot be intuitively weighed by the state cost matrix. The Modified Rodrigues Parameters (MRP), also known as conformal rotation vector, are a stereographic projection of a unit quaternion $\mathbf{q} \in \mathbb{H}$ onto the pure imaginary hyperplane. They offer a convenient minimal representation of rotation with good convergence [45] and allow weighing the axis errors individual. The MRP transformation is defined by:

$$\text{mrp} : \mathcal{X} \rightarrow \mathbb{R}^{12}, \quad x \mapsto \left[\mathbf{p}, \frac{\mathbf{q}_{i,j,k}}{1 + \mathbf{q}_r}, \mathbf{v}, \boldsymbol{\omega} \right]^\top. \quad (64)$$

Using the MRPs, the state error $e_k \in \mathcal{E} \subseteq \mathbb{R}^{12}$ is defined as

$$e_k = \text{mrp}(x_k \ominus x_{\text{ref},k}). \quad (65)$$

Two methods were compared for the control. The optimal *linear quadratic regulator* (LQR) and *nonlinear model predictive control* (NMPC)

3.4.2. Linear Quadratic Regulator

The linear quadratic regulator is the optimal controller for linear time invariant systems. The model described in Section 2.1 is non-linear and must be approximated by a linear model.

3.4.2.1. Linearization

The error of the state and control to the reference can be described using the operator of Equation 63 as

$$\tilde{\mathbf{x}}(t) = \mathbf{x}(t) \ominus \mathbf{x}_{\text{ref}}(t) \quad (66)$$

$$\tilde{\mathbf{u}}(t) = \mathbf{u}(t) - \mathbf{u}_{\text{ref}}(t). \quad (67)$$

The error state shall use MRP as a orientation representation by transforming the states into error state $\mathcal{E}$ as follows:

$$\text{mrp}(\tilde{\mathbf{x}}(t)) = \text{mrp}(\mathbf{x}(t) \ominus \mathbf{x}_{\text{ref}}(t)) \in \mathcal{E}. \quad (68)$$

Assuming the error will be small - so the state is always close to the reference - the MRP function can be approximated as a linear function allowing

$$\text{mrp}(\tilde{\mathbf{x}}(t)) = \text{mrp}(\mathbf{x}(t)) \ominus \text{mrp}(\mathbf{x}_{\text{ref}}(t)). \quad (69)$$

The states are renamed to

$$\tilde{\mathbf{x}}_{\mathcal{E}}(t) = \mathbf{x}_{\mathcal{E}}(t) - \mathbf{x}_{\mathcal{E},\text{ref}}(t) \quad (70)$$

and the dynamic function to

$$\dot{\mathbf{x}}_{\mathcal{E}}(t) = \text{mrp}(\mathbf{f}(\text{mrp}^{-1}(\mathbf{x}_{\mathcal{E}}(t)), \mathbf{u}(t))) := \mathbf{f}_{\mathcal{E}}(\mathbf{x}_{\mathcal{E}}(t), \mathbf{u}(t)) \quad (71)$$

for better readability. The dynamic function can be linearized by developing the first order Taylor expansion at $(\mathbf{x}_{\mathcal{E},\text{ref}}(t), \mathbf{u}(t))$:

$$\begin{aligned} \mathbf{f}_{\mathcal{E}}(\mathbf{x}_{\mathcal{E}}(t), \mathbf{u}(t)) \approx & \mathbf{f}_{\mathcal{E}}(\mathbf{x}_{\mathcal{E},\text{ref}}(t), \mathbf{u}_{\text{ref}}(t)) + \\ & + \underbrace{\frac{\partial \mathbf{f}_{\mathcal{E}}}{\partial \mathbf{x}_{\mathcal{E}}}}_{\mathbf{A}(t)} \bigg|_{(\mathbf{x}_{\text{ref}}(t), \mathbf{u}_{\text{ref}}(t))} \mathbf{x}_{\mathcal{E}}(t) + \underbrace{\frac{\partial \mathbf{f}_{\mathcal{E}}}{\partial \mathbf{u}}}_{\mathbf{B}(t)} \bigg|_{(\mathbf{x}_{\text{ref}}(t), \mathbf{u}_{\text{ref}}(t))} \mathbf{u}(t) \end{aligned} \quad (72)$$

The time derivative of Equation 67 leads to

$$\dot{\tilde{\mathbf{x}}}_{\mathcal{E}}(t) = \dot{\mathbf{x}}_{\mathcal{E}}(t) - \dot{\mathbf{x}}_{\mathcal{E},\text{ref}}(t) \quad (73)$$

and assuming the reference trajectory is respecting the dynamics can be substituted to

$$\dot{\tilde{\mathbf{x}}}_{\mathcal{E}}(t) = \dot{\mathbf{x}}_{\mathcal{E}}(t) - \mathbf{f}_{\mathcal{E}}(\mathbf{x}_{\mathcal{E},\text{ref}}(t), \mathbf{u}_{\text{ref}}(t)). \quad (74)$$

With the approximated Taylor Equation 72, the non-linear reference dynamics cancel out resulting to the linear time variant system:

$$\dot{\mathbf{x}}_{\mathcal{E}}(t) = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{B}(t)\mathbf{u}(t). \quad (75)$$

To obtain a time invariant system, we define representative operating point $\mathbf{x}_{\text{op}}, \mathbf{u}_{\text{op}}$, where the vessel experiences no acceleration or angular acceleration. Finding such an operating point is the same as finding the root of the dynamic functions' $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$ second order terms. The root finding can be formulated as the optimization problem

$$\underset{\mathbf{u}_{\text{op}}}{\text{argmin}} \mathbf{f}(\mathbf{x}_{\text{op}}, \mathbf{u}_{\text{op}})^{\top} \mathbf{W} \mathbf{f}(\mathbf{x}_{\text{op}}, \mathbf{u}_{\text{op}}) \quad (76)$$

with $\mathbf{W} = \text{diag}([0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1])$ to find the inputs $\mathbf{u}_{\text{op}}$ at state $\mathbf{u}_{\text{op}}$. This results in the time invariant dynamic matrix and input matrix

$$\mathbf{A} = \left. \frac{\partial \mathbf{f}_{\mathcal{E}}}{\partial \mathbf{x}_{\mathcal{E}}} \right|_{\mathbf{x}_{\text{op}}, \mathbf{u}_{\text{op}}}, \mathbf{B} = \left. \frac{\partial \mathbf{f}_{\mathcal{E}}}{\partial \mathbf{u}} \right|_{\mathbf{x}_{\text{op}}, \mathbf{u}_{\text{op}}}. \quad (77)$$

3.4.2.2. Controller Synthesis

Due to the linearization at the reference state, the LQR performs best when the system state $\mathbf{x}(t)$ is very close to the reference state $\mathbf{x}_{\text{ref}}(t)$. The position, velocity and orientation are referenced in world space $\mathcal{W}$, while the angular velocity lives in fixed body frame $\mathcal{B}$. If for example the system is linearized at a state where the vessel is heading north and then operating heading west, the error states referenced in fixed world coordinates are very large, even though the dynamics are independent of the heading (yaw angle). It makes sense to carry out a coordinate transformation $\mathbf{T}_{\mathcal{W}}^{\mathcal{B}'} : \mathcal{X} \rightarrow \mathcal{X}$ of the error state from world coordinates into body coordinates. The body fixed state $\mathbf{x}^{\mathcal{B}'}$ at current state $\mathbf{x}$ is specified by

$$\mathbf{x}^{\mathcal{B}'} = \mathbf{T}_{\mathcal{W}}^{\mathcal{B}'}(\mathbf{x}) = \begin{pmatrix} \mathbf{q} \odot \mathbf{p} \\ \mathbf{q}^{-1} \\ \mathbf{q} \odot \mathbf{v} \\ \hat{\boldsymbol{\omega}} \end{pmatrix}. \quad (78)$$

The control input using the coordinate transformation and input reference $\mathbf{u}_{\text{ref}}$ is

$$\mathbf{u} = \mathbf{u}_{\text{ref}} - \mathbf{K} \text{mrp}(\mathbf{T}_{\mathcal{W}}^{\mathcal{B}'}(\mathbf{x}) - \mathbf{T}_{\mathcal{W}}^{\mathcal{B}'}(\mathbf{x}_{\text{ref}})). \quad (79)$$

This results to the discrete policy:

$$\pi(\mathbf{x}_k, \mathbf{x}_{\text{ref},k}, \mathbf{u}_{\text{ref},k}) = \mathbf{u}_{\text{ref},k} - \mathbf{K} \text{mrp}(\mathbf{T}_{\mathcal{W}}^{\mathcal{B}'}(\mathbf{x}_k) - \mathbf{T}_{\mathcal{W}}^{\mathcal{B}'}(\mathbf{x}_{\text{ref},k})). \quad (80)$$

The weight matrix $\mathbf{Q}$ and $\mathbf{R}$ are tuned to a good compromise of regulating speed versus the control effort.

3.4.3. Model Predictive Control

The model predictive controller computes input $\mathbf{u}^*$ given the current state $\mathbf{x}_k$ and a state and input reference trajectories computed by the motion planner. This can be written as the policy:

$$\mathbf{u}^* = \pi_{\text{mpc}}(\mathbf{x}_k, \mathbf{x}_{\text{ref}}, \mathbf{u}_{\text{ref}}). \quad (81)$$

The controller is designed for a finite horizon N . For each step a state input $\tilde{\mathbf{u}}_n \forall n \in \{0, \dots, N\}$ is defined. The corresponding states are $\tilde{\mathbf{x}}_n \forall n \in \{0, \dots, N+1\}$. For these states the cost function

$$\begin{aligned} J_{\text{mpc}} = & \sum_{n=0}^N \text{mrp}(\tilde{\mathbf{x}}_n \ominus \mathbf{x}_{\text{ref},k+n})^\top \mathbf{Q}_{\text{mpc}} \text{mrp}(\tilde{\mathbf{x}}_n \ominus \mathbf{x}_{\text{ref},k+n}) \\ & + \sum_{n=0}^N (\tilde{\mathbf{u}}_n - \mathbf{u}_{\text{ref},n+k})^\top \mathbf{R}_{\text{mpc}} (\tilde{\mathbf{u}}_n - \mathbf{u}_{\text{ref},n+k}) \\ & + (\tilde{\mathbf{x}}_{N+1} \ominus \mathbf{x}_{\text{ref},k+N+1})^\top \mathbf{Q}_{T,\text{mpc}} \text{mrp}(\tilde{\mathbf{x}}_{N+1} \ominus \mathbf{x}_{\text{ref},k+N+1}) \end{aligned} \quad (82)$$

is defined. It penalizes the error to the reference actions and states using the weight matrices $\mathbf{Q}_{\text{mpc}}$ and $\mathbf{R}_{\text{mpc}}$ and terminal state weight $\mathbf{Q}_{T,\text{mpc}}$.

The optimal input sequence $\tilde{\mathbf{u}}$ is determined with the following optimization problem. The initial condition is contained to the current state value $\mathbf{x}_k$. The states are contained to the dynamics using Euler integration with time step δt_{mpc} . To ensure the right rotation properties of the state quaternion $\mathbf{q}$, it must be a normalized unit quaternion for every time step:

$$\begin{aligned} \underset{\mathbf{u}_{0:N}}{\text{argmin}} J_{\text{mpc}} \quad \text{s.t.} \quad & \tilde{\mathbf{x}}_0 = \mathbf{x}_k \\ & \tilde{\mathbf{x}}_{n+1} = \tilde{\mathbf{x}}_n + \mathbf{f}(\tilde{\mathbf{x}}_n, \tilde{\mathbf{u}}_n) \cdot \Delta t_{\text{mpc}} \quad \forall n \in \{0, \dots, N\} \\ & \mathbf{q}_n = \frac{\mathbf{q}_n}{\|\mathbf{q}_n\|} \quad \forall n \in \{0, \dots, N\} \\ & u_{\min,i} \leq \tilde{u}_{n,i} \leq u_{\max,i} \quad \forall n \in \{0, \dots, N\} \quad \forall i \in \{0, \dots, p\} \end{aligned} \quad (83)$$

3.4.4. Multi Robot Extension

The model predictive controller has the advantage that it can easily be extended to avoid collision in a multi robot environment. For a known preplanned second robot trajectory $\mathbf{x}'_{\text{ref}}$ the collision constraint

$$\|\mathbf{p}_{\text{ref},n+k} - \mathbf{p}'_{\text{ref},n+k}\| < 2r \quad \forall k \in \{0, \dots, N\} \quad (84)$$

is added. On other approach is to propagate all robots within the same optimization problem making all robots input trajectories decision variables.

4. Experiments

4.1. Model Implementation

The dynamics described in Section 2.1 are implemented in Rust, which, as a high-performance compiled language, enables real-time modeling, while the nalgebra library provides efficient tools for linear algebra computations. All model parameters shown in Table 3 in the Appendix are read from a configuration file.

4.1.1. Differential Equation

The forward propagation of the system dynamics function $\mathbf{f}$ is approximated with the *explicit Euler method* for solving ordinary differential equations. The equation for step $k + 1$ depending on step i

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k) \cdot \Delta t_\varepsilon, \quad (85)$$

describes the state discrete propagation for Δt_ε time step. In the implementation an update rate of 1000Hz is chosen equating to $\Delta t_\varepsilon = 0.001$.

4.1.2. Aerodynamic Coefficients

The dynamic function described in Section 2.1 relies on the fluid-dynamic coefficients $C_D(\alpha, \text{Re})$ and $C_L(\alpha, \text{Re})$, which must be determined for accurate modeling. XFOIL, a well-established analysis tool for subcritical airfoils, is employed to compute these aerodynamic coefficients [46]. However, direct use of XFOIL within the dynamic simulation is computationally inefficient, as the solving process is relatively slow.

Since the Reynolds number Re depends on velocity but exhibits only small changes to the coefficient within the considered range, it can be assumed constant near operating velocity for practical purposes. The dependence on the angle of attack α is captured by fitting a polynomial to discrete samples obtained from XFOIL.

The polynomial fit of order n is defined by the coefficients $\mathbf{C} \in \mathbb{R}^n$ and

$$C(\alpha) = \sum_{i=0}^n C_i \alpha^i. \quad (86)$$

To illustrate this approach, Figure 5 presents the XFOIL results for C_D and C_L as functions of α at three different Reynolds numbers, alongside the corresponding polynomial interpolations. The resulting coefficient for the lift surfaces and masts are collected in Table 2

For very large angles of attack larger of $\alpha \gg 30^\circ$ the foil stalls and all lift is lost, but this range the error of the polynomial fit is very large. Multiplying by the sigmoid function

$$C''(\alpha) = \frac{1}{1 + e^{50(-30-\alpha)}} \frac{1}{1 + e^{50(-30+\alpha)}} \cdot C(\alpha). \quad (87)$$

is a differentiable way to map large α to zero.

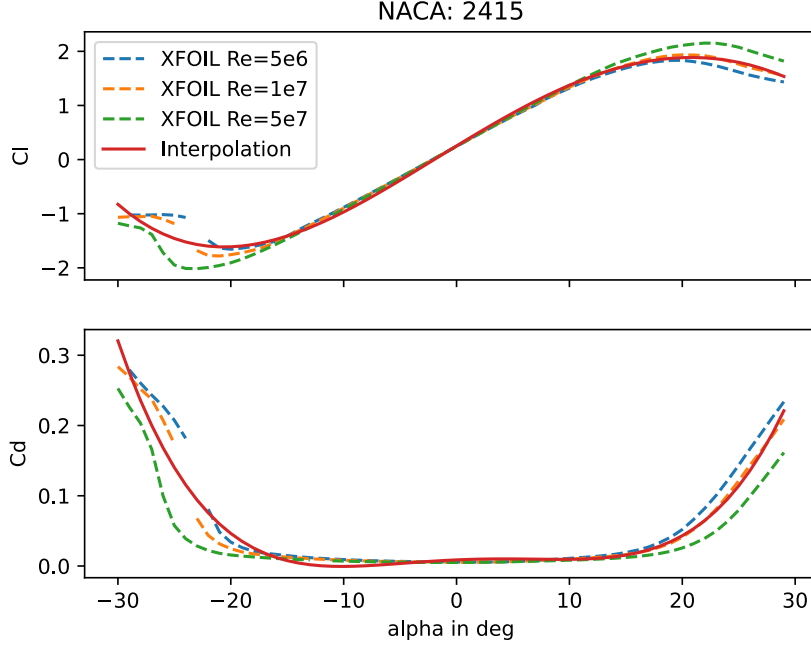


Figure 5: Comparison XFOIL results with polynomial interpolations.

Coefficient	Interpolation Coefficients
$C_{L,\text{lift}}$	$[7.09 \cdot 10^{-7}, -9.91 \cdot 10^{-5}, -5.77 \cdot 10^{-4}, 1.27 \cdot 10^{-1}, 2.53 \cdot 10^{-1}]$
$C_{D,\text{lift}}$	$[4.42 \cdot 10^{-7}, -1.95 \cdot 10^{-6}, -5.77 \cdot 10^{-4}, 1.27 \cdot 10^{-1}, 2.53 \cdot 10^{-1}]$
$C_{L,\text{mast}}$	$[-2.47 \cdot 10^{-7}, -1.14 \cdot 10^{-4}, 1.47 \cdot 10^{-4}, 1.31 \cdot 10^{-1}, -0.678 \cdot 10^{-3}]$
$C_{D,\text{mast}}$	$[6.03 \cdot 10^{-7}, 7.36 \cdot 10^{-7}, -1.30 \cdot 10^{-4}, -2.94 \cdot 10^{-4}, 9.01 \cdot 10^{-3}]$

Table 2: Interpolation results of the aerodynamic coefficients

4.1.3. Qualitative Validation

The implemented model supports real-time simulation with controller inputs depicted in Figure 6, which enables qualitative validation against a remotely controlled 1:3 scale model depicted in Figure 1. A logging system was implemented that records data at 100 Hz into a database using sensor fusion from a 9-axis IMU, GPS, and dual sonar sensors. However, parameter estimation and model fitting based on these measurements are beyond the scope of this thesis.

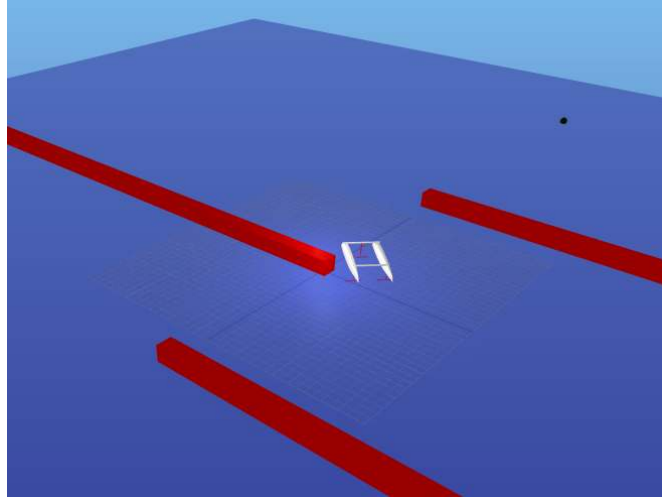


Figure 6: Computer simulation of the vessel on its course.

4.2. Motion Planning

4.2.1. Implementation

For motion planning the implementation of RRT and SST by the Open Motion Planning Library (OMPL) are used [47]. For implementation of the state metric, constraints and collision checking C++ is chosen, for good performance interfacing with OMPL. The state propagator uses C bindings to the Rust model described in Section 4.1.

4.2.2. Test Cases

To find out whether motion planning is possible at all, and if so, to quantitatively evaluate of the implemented motion planning methods, several test cases were designed. These test cases should cover a variety of situations that challenge the planning algorithms. For all test cases the workspace is constrained to an 80 meter by 80 meter square. As the most simple test case the vessels has a start position where it is heading at the goal region and must travel 40 meters in a *straight* line. The planner must comply with the dynamics and state limits. In the next test case, a spherical *obstacle* of diameter 30cm representing a buoy is introduced at the midpoint between start position and goal region. In this test case obstacle avoidance is challenged. The third test is much more constrained, forcing the vehicle on a *slalom* trajectory with box obstacles to reach the goal region. As a last benchmark the ability to change heading as fast as possible is tested. The vessels starting configuration heads to the opposite direction of the goal region enforcing a *U-turn*. Figure 7 shows the test cases with example trajectories.

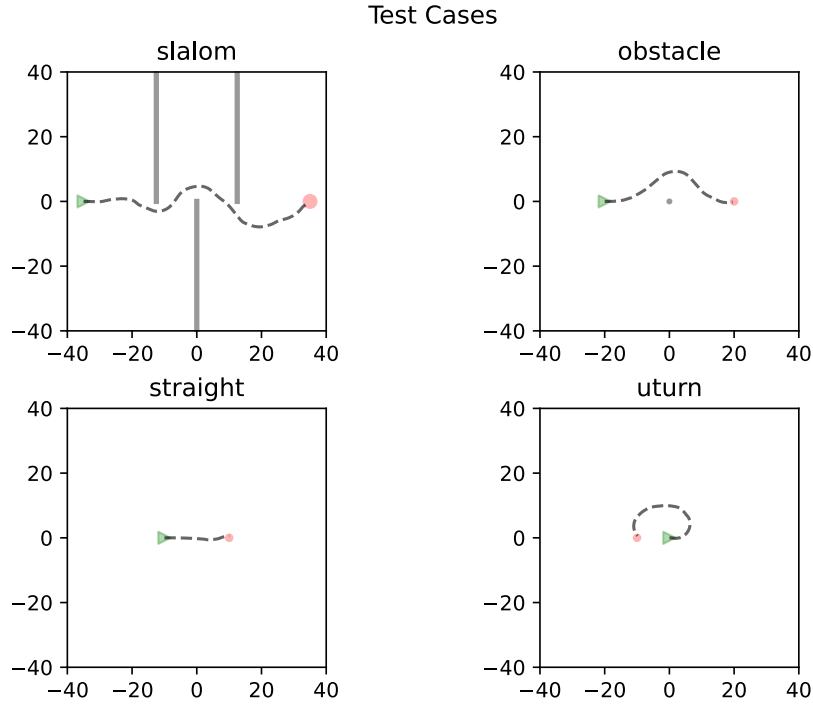


Figure 7: Test cases with exemplary results.

4.2.3. Results

Running each test 50 times with RRT and SST reveals that motion planning for these hydrofoil dynamics is possible, despite the highly complex dynamics and very tight configuration constraints both algorithms find solutions. Figure 8 shows the success rate of RRT and SST for the different test cases.

Both algorithms have a higher success rate for the simpler tests *straight* and *obstacle*. For the more complex slalom problem the success rate drops, and for the U-turn no run of the 50 SST executions could find a solution.

The RRT algorithm has for every test case a higher chance of finding a solution in the 60 second compute time than SST. This may happen, because RRT does no pruning and selecting which makes it computationally less expensive per sample than SST. Therefore, RRT has greater chance of finding a path due to conduction more samples in the same compute time.

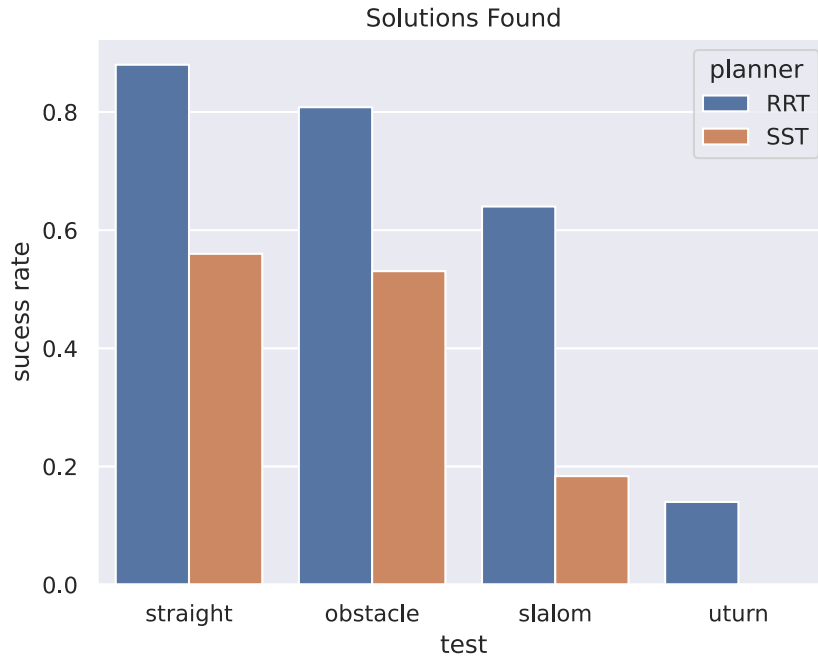


Figure 8: Success rate of both planners for the various test cases.

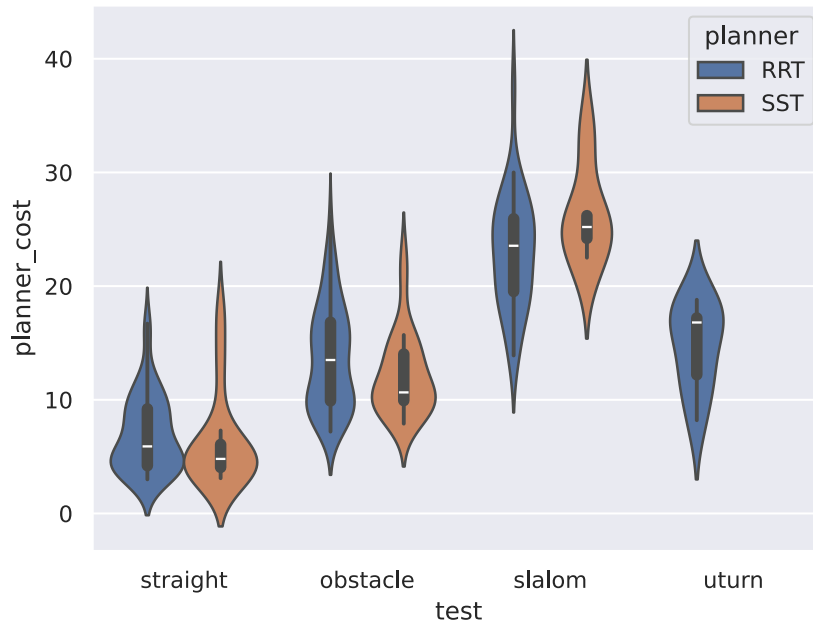


Figure 9: Performance of both plannerd for the various test cases.

Figure 9 compares the cost - which is the trajectory length with respect to time - of all successfully planned trajectories. The distribution of the solutions cost for RRT and SST for the four test cases are depicted. Here the advantages of SST can be seen. SST delivers a lower mean solution time for the straight and obstacle

test case. Here SST's approximate asymptotic optimality may be the reason for the better performance compared to RRT. At the more complex test RRT does not only have a higher success rate but also delivers trajectory with a better quality regarding cost. The relatively short compute time may favor RRT.

Looking at the sampled states of a successful RRT and an unsuccessful SST run in Figure 10 reveals the large difference in sampled states.

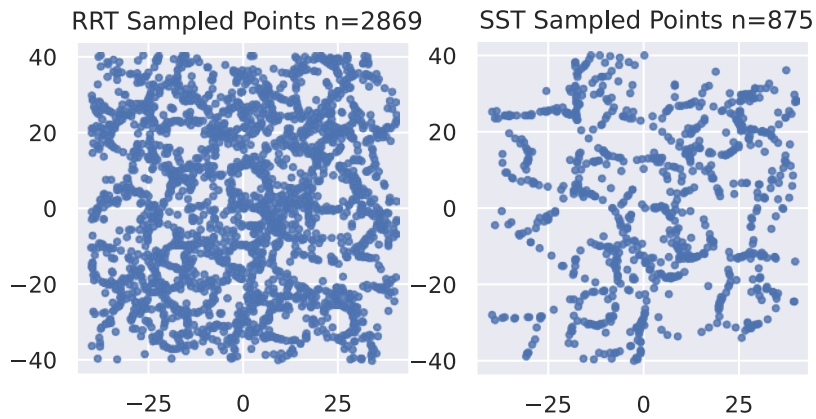


Figure 10: Comparison of sampled state of both planners.

Next the influence of changing the computation time is examined. The slalom problem is solved again with RRT and SST giving the algorithms a computation time of 60s, 120s and 600s. Figure 11 shows that the success rate of both algorithms increases. SST improves disproportionately better than RRT.

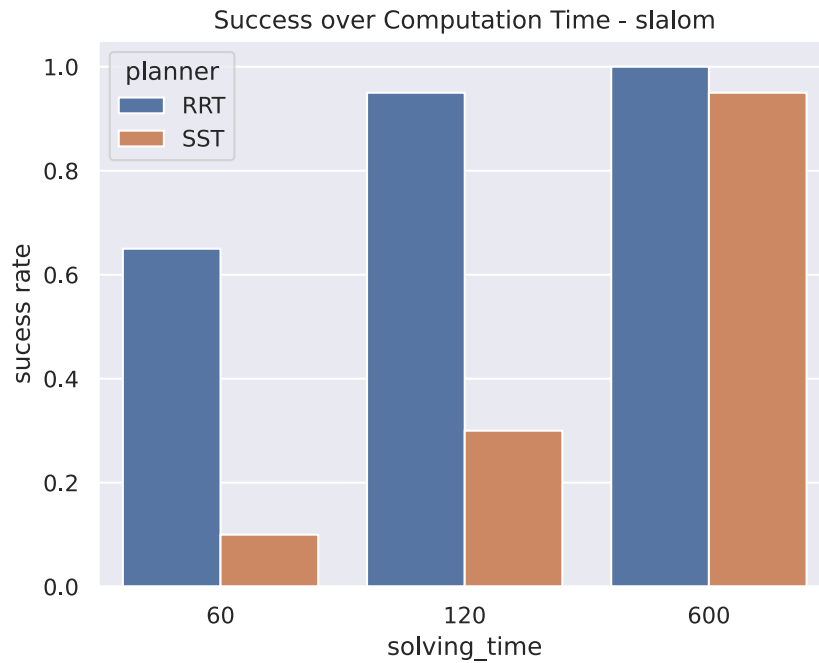


Figure 11: Planner success rate with different computation time.

Figure 12 shows that the quality of the plans improves with larger runtime. SST delivers a faster average trajectory for each computation time length.

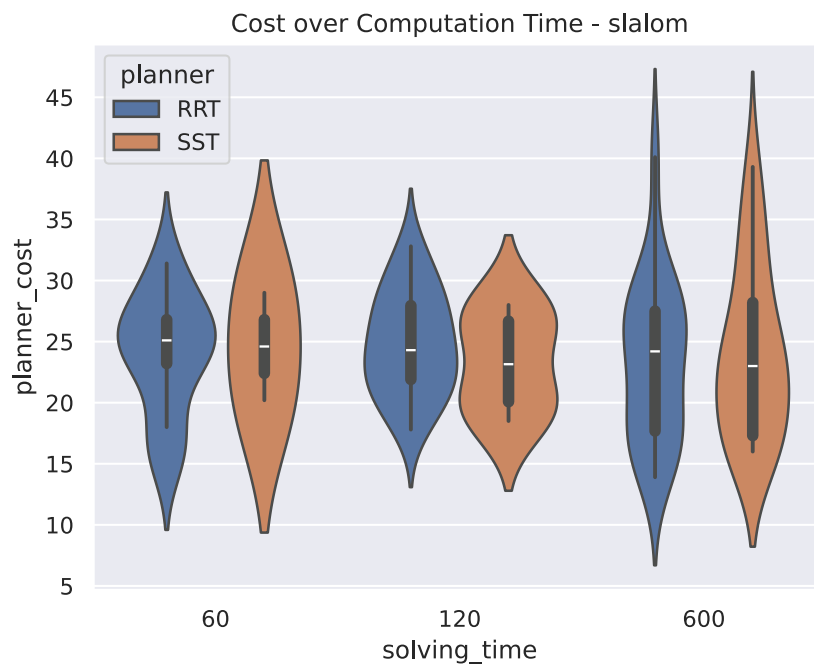


Figure 12: Planner quality with different computation time.

Looking at the best slalom trajectory at Figure 13 of the data set shows that it is still far from optimal. This plan was done by SST. The vehicle could fly closer to the obstacles and approach the goal region from a better angle.

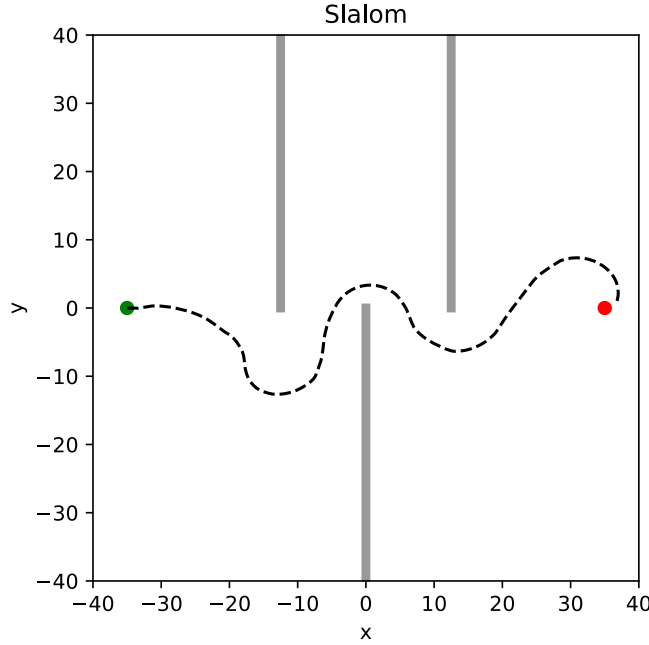


Figure 13: Best result for the slalom trajectory.

4.3. Control

4.3.1. Implementation

For LQR the model dynamics need to be linearized by evaluating Jacobians at an operating state. The solver for the MPC relies on the Jacobian and hessian of the cost function. Therefore, the model dynamics are reimplemented with a computer algebra system enabling auto differentiation. In this approach Casadi with its Python API is used [48]. The model parameters are read from the same configuration file as described in Section 4.1. A propagation test ensures that the Rust and CasADi model implementation behave exactly the same.

4.3.1.1. LQR

For the LQR, the error dynamics of Section 3.4.1 are implemented. CasADi's jacobian function then is used to obtain the dynamic matrix $\mathbf{A}$ and $\mathbf{B}$ at a carefully chosen operating state. The python library *python control* implements a solver for the Riccati differential equation leading to the desired control matrix $\mathbf{K}$. For an operating state and action the optimization problem of Equation 76 is implemented and solved with IPOPT [49]. The chosen acceleration free state with only forward velocity

$$\mathbf{x}_{\text{op}} = \begin{pmatrix} [0, 0, 0]^\top \\ [1, 0, 0, 0]^\top \\ [5, 0, 0]^\top \\ [0, 0, 0]^\top \end{pmatrix} \quad (88)$$

solved to the steady action

$$\mathbf{u} = \begin{pmatrix} -1.44 \\ -1.44 \\ -1.44 \\ 0.0 \\ 520.713 \end{pmatrix}. \quad (89)$$

4.3.1.2. MPC

The cost function of the MPC is implemented using the CasADi Python API. CasADi provides class `Opti` for solving non-linear programs with IPOPT. The Runtime of the solver is improved by using the code generation and just in time compilation capabilities of CasADi. The values of the input vector $\mathbf{u}$ are of different orders. To ensure good conditioning of the optimization problem the inputs are scaled by a constant. As a time step size $\Delta t_{\text{mpc}} = 0.1$ is chosen. For larger steps the Euler integration starts to get unstable.

4.3.2. Test Environment

The disturbed initial state is $\hat{\mathbf{x}}_0 \neq \mathbf{x}_{\text{ref},0}$. The initial state is disturbed by bias free Gaussian noise with

$$\boldsymbol{\sigma} = \begin{pmatrix} \boldsymbol{\sigma}_p \\ \boldsymbol{\sigma}_q \\ \boldsymbol{\sigma}_v \\ \boldsymbol{\sigma}_\omega \end{pmatrix} \in \mathbb{R}^{12}, \quad \boldsymbol{\sigma}_p, \boldsymbol{\sigma}_q, \boldsymbol{\sigma}_v, \boldsymbol{\sigma}_\omega \in \mathbb{R}^3 \quad (90)$$

with covariance matrix $\boldsymbol{\Sigma} = \text{diag}(\boldsymbol{\sigma}) \in \mathbb{R}^{12 \times 12}$.⁹ The randomly sampled initial state then is

$$\hat{\mathbf{x}}_0 = \begin{pmatrix} \mathbf{p} + \boldsymbol{\eta}_p \\ \text{quat}(\boldsymbol{\eta}_q) \odot \mathbf{q} \\ \mathbf{v} + \boldsymbol{\eta}_v \\ \boldsymbol{\omega} + \boldsymbol{\eta}_\omega \end{pmatrix} \text{ with } \boldsymbol{\eta} \sim \mathcal{N}(0, \boldsymbol{\Sigma}) \quad (91)$$

adding noise to each state variable in Euclidean vector space. For the orientation we denote

$$\text{quad} : \mathbb{R}^3 \rightarrow \mathbb{H}, \quad \boldsymbol{\theta} \mapsto \begin{pmatrix} \cos\left(\frac{\|\boldsymbol{\theta}\|}{2}\right) \\ \frac{\boldsymbol{\theta}}{\|\boldsymbol{\theta}\|} \sin\left(\frac{\|\boldsymbol{\theta}\|}{2}\right) \end{pmatrix} \quad (92)$$

as the mapping from axis-angle vectors to unit quaternion.

⁹The rotation noise vector $\boldsymbol{\eta}_q$ defines the noise as axis angle rotation.

4.3.3. LQR Parameters Tuning

To tune the behavior of the Linear Quadratic Controller is defined by the state cost matrix $\mathbf{Q}_c$ and input cost matrix $\mathbf{R}_c$ are determined. $\mathbf{Q}_c \in \mathbb{R}^{12 \times 12}$ and $\mathbf{R}_c \in \mathbb{R}^{5 \times 5}$ are diagonal matrices, weighing each state and input independently that makes a total of 17 parameters. In this section a set of reasonable weight with good performance are searched with trial and error. So intuition and a sense for the dynamics is needed.

4.3.3.1. Attitude

The orientation of the vessel is very important for the trajectory following since the dynamic forces are very dependent on the angle of attack and sideslip in the water. A deviation in orientation guarantees integrating a positional error. So, at first the parameters are adjusted such that the controller creates a good response to a disturbance in yaw, pitch and roll. A deviation of 0.1 rad ($\approx 5.7^\circ$) should be compensated in as fast as possible. Figure 14 shows the error of each state and inputs over a five-second steady state trajectory. The initial state has a deviation in pitch angle φ of 0.1 rad. $\mathbf{Q}_c$ and $\mathbf{R}_c$ were tuned such that the error of φ converges to zero in 1-2 seconds. The resulting actions on the foils can be seen in the lower left plot. To compensate for the positive pitch error the front foils are pitched up while the aft foil is pitched down. Roll and yaw are tuned in the same manner.

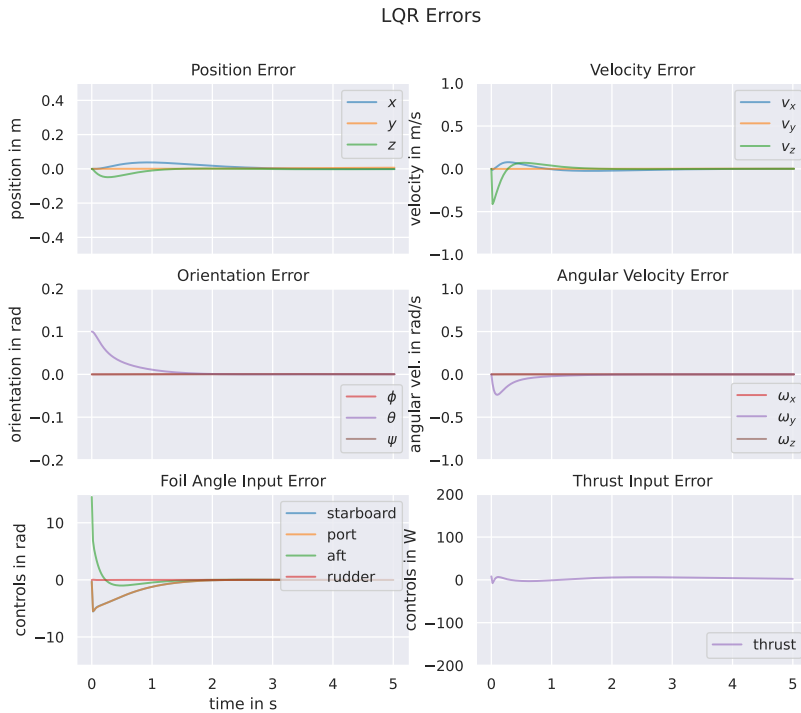


Figure 14: LQR performance for disturbed pitch angle.

4.3.3.2. Height

For the hydrofoil vessel the ride height is very important. If the riding height is too high and the foil ventilate losing all lift and being too low submerges the hulls creating drag. For the z -state a larger weight in the cost function is chosen to trigger fast responses to small height errors. An example disturbance is depicted in Figure 15.

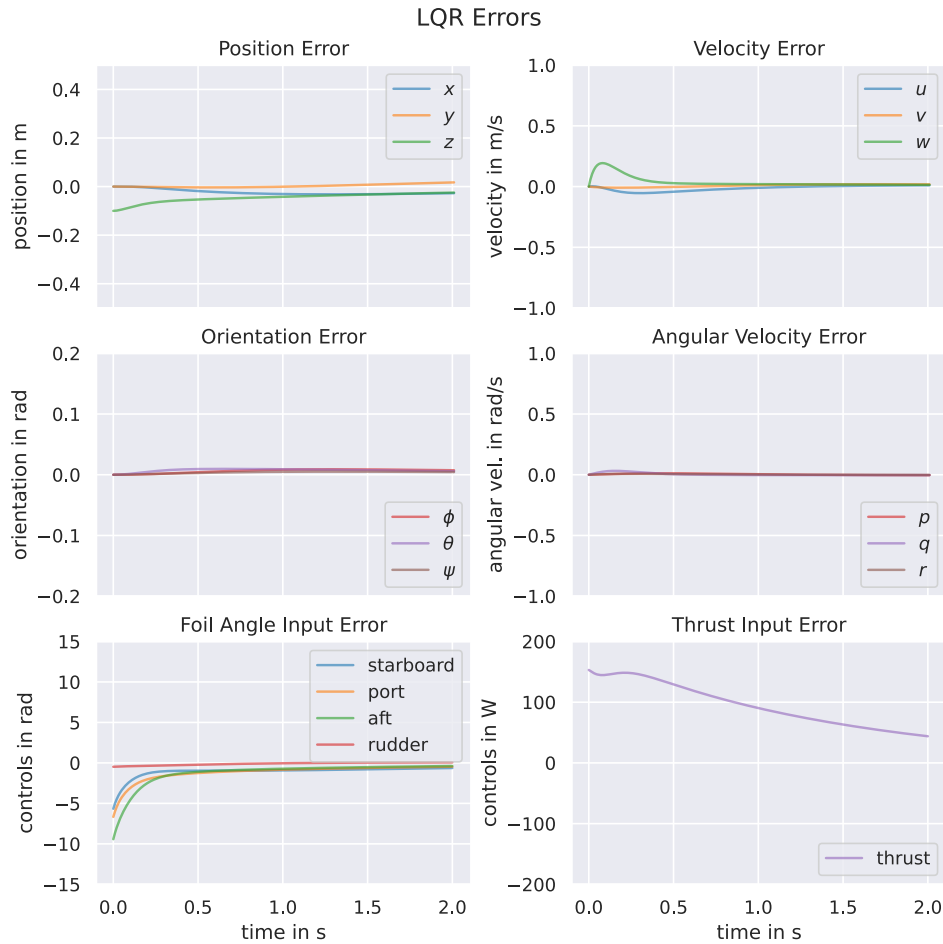


Figure 15: LQR performance on height disturbance.

4.3.3.3. Longitudinal Error

The longitudinal error is coupled to the thrust input. For this axis oscillations occurred while tuning. If the state weight on x is large the vessel over compensates with the acceleration and overshoots the target position. Increasing the state cost for u - the velocity in x -direction - to 0.1 has a dampening effect, mitigating the overshoot. An example trajectory for disturbance in x is shown in Figure 16.

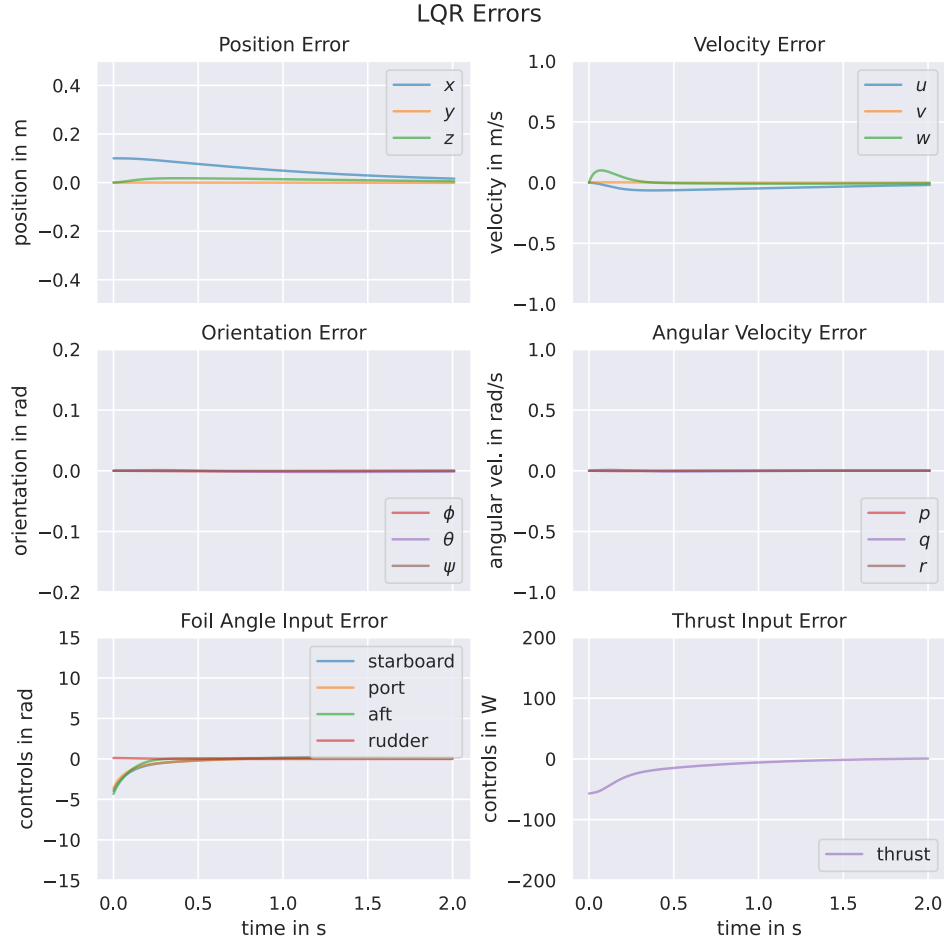


Figure 16: LQR performance on longitudinal disturbance.

4.3.3.4. Lateral Error

Tuning the LQRs cost function for lateral error in the y direction turns out to be more challenging. Figure 17 shows a tuned LQR trying to compensate a 0.1m lateral deviation of the initial condition. The controller is unstable and increasingly deviates from the reference trajectory. This effect appears due to the non-linearity of the dynamics. For stability the controller should create a positive rudder input for a positive lateral error, to steer back onto the trajectory. Examining the y coordinates of the linearized dynamics

$$\dot{\tilde{\mathbf{x}}}_{\mathcal{E}} = \mathbf{A}\tilde{\mathbf{x}}_{\mathcal{E}} + \mathbf{B}\tilde{\mathbf{u}} \quad (93)$$

with $\mathbf{x}_{\mathcal{E}} = [0, \tilde{y}, 0, 0, 0, 0, 0, 0, 0, 0, 0]$ and $\tilde{\mathbf{u}} = [0, 0, 0, u_{\text{rudd}}, 0]$ reveals $\ddot{y} = 0.991 \cdot u_{\text{rudd}}$. A positive rudder action accelerates the lateral error, the opposite of the expected behavior. Taking a closer look at Figure 17 shows that the lateral error is reduced for $\approx 0.5s$ before increasing. The change in yaw due to positive rudder action induces a side-slip angle on the forward masts, that then dynamically create lift the negative lateral error direction. After a short time the velocity in the new direction increases, reducing the side-slip angle and

therefore then increasing the lateral error. The only option for the LQR design is to weight the lateral error in Q_c with zero.

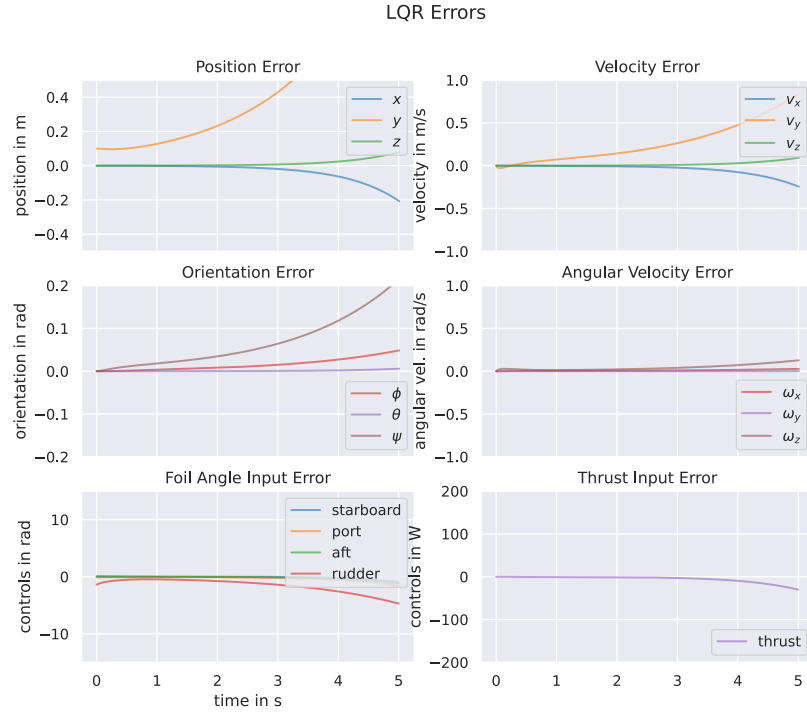


Figure 17: LQR performance on lateral disturbance.

The developed controller with

$$\begin{aligned} \mathbf{Q} &= \text{diag}[0.8, 0.0, 4.0, 5.0, 10.0, 5.0, 0.1, 0.0, 0.1, 0.001, 0.001, 0.001] \\ \mathbf{R} &= \text{diag}[(10^{-4}, 10^{-4}, 10^{-4}, 10^{-4}, 10^{-6})] \end{aligned} \quad (94)$$

can be tested with the planned trajectories from Section 4.2.3. Figure 18 shows the reference and simulated trajectories with an initial error of $\tilde{x}_{\mathcal{E},0} = [0.0, 0.1, 0.05, 0.0, 0.07, 0.0, 0.0, 0, 0, 0, 0, 0]$. The controller can follow the complex trajectories with acceptable error.

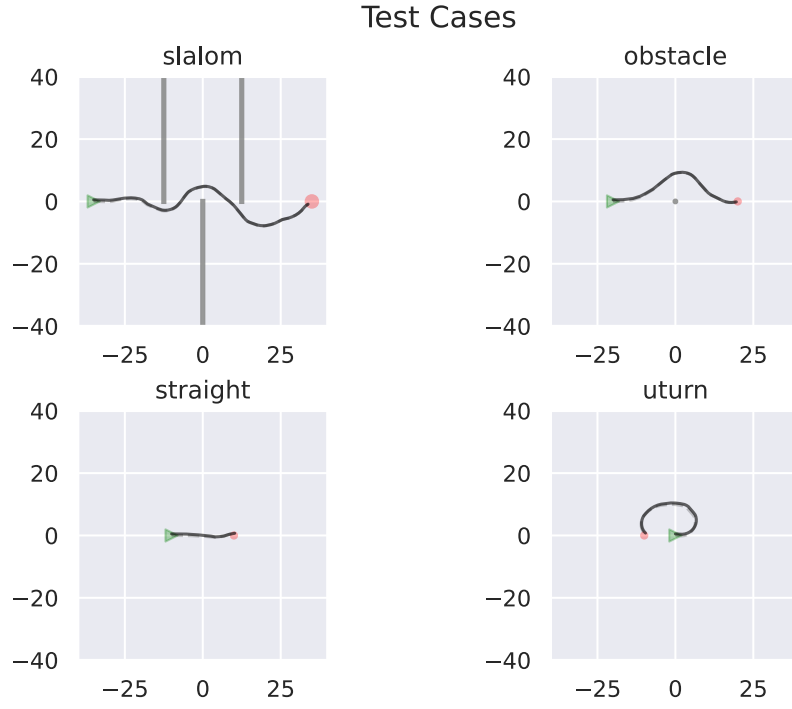


Figure 18: Top view on four motion planning references.



Figure 19: Positional error distribution.

Figure 19 depicts the error distribution of the fastest ten motion planning trajectories for each test with initial state disturbance normal noise of $\sigma = 0.1$ for first order states and $\sigma = 0.05$ for second order states. The plot shows that disturbances in x and z direction are reduced by two orders of magnitude, indicating a

well-tuned controller. As expected, the linear controller struggles to compensate disturbances in lateral y direction. The error is one order of magnitude larger than the other two dimensions. The average y -error is close to the standard distribution of the initial disturbance, suggesting that there is no compensation of the error in y direction at all.

4.3.4. MPC Parameter Tuning

The challenging local non-linear instability described in the previous section is a case where the Non-linear Model Predictive Controller poses a possibility for stability. With a well sized horizon and good cost function the controller may present a stable solution.

With a horizon of $N = 30$ steps at $\Delta t_{\text{mpc}} = 0.01$ resulting in a 300ms time frame the prediction horizon captures the non-linear dynamics for lateral tracking. The cost function must reward a choosing larger short term penalty in y and yaw to archive a long term improvement. This is archived by decreasing the cost for lateral error in state weight matrix $\mathbf{Q}$ and forcing a long term improvement by increasing it in the terminal state weight $\mathbf{P}$. Figure 20 shows an example trajectory where a lateral disturbance is successfully compensated by the MPC. The error in y is increasing for a few hundred milliseconds while adjusting the heading, and then converging to zero.

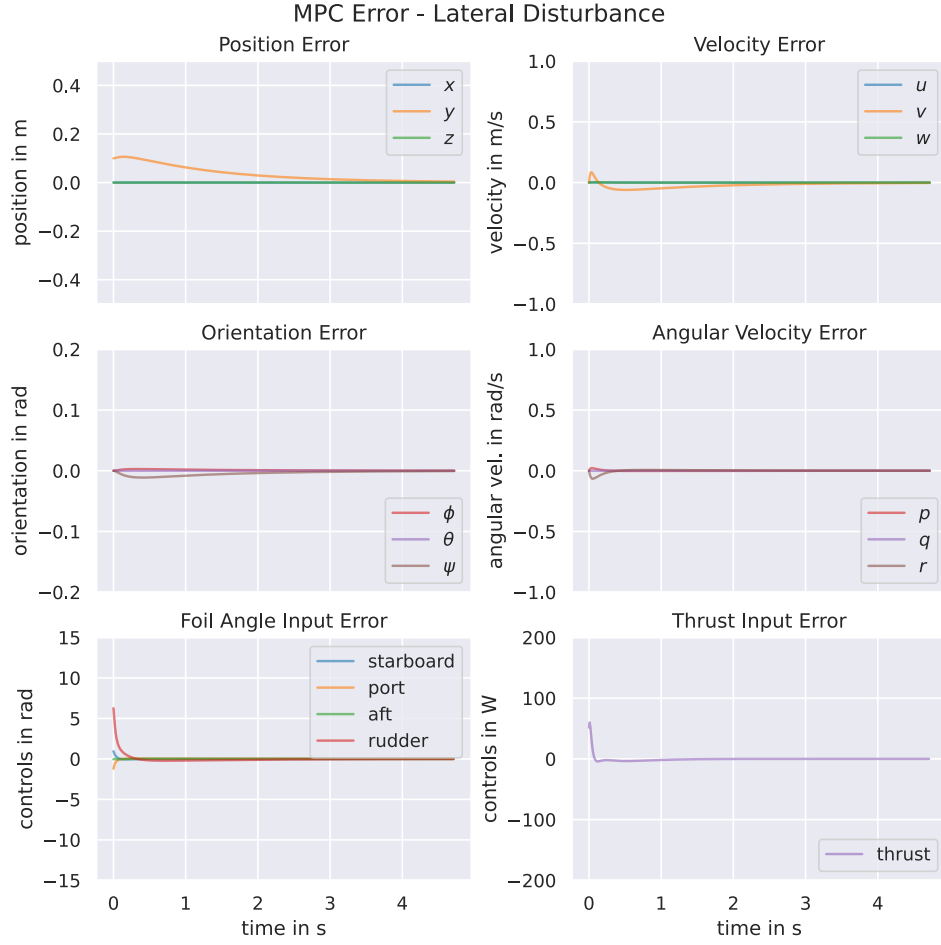


Figure 20: Lateral stability of MPC.

Unfortunately due to issues in the implementation the model predictive controller could not be tested on any reference trajectory from the motion planner. For unknown reasons the solver failed to converge at arbitrary points. Therefore, the following benchmarks have been run only on selected trajectories.

Figure 21 shows the average error distribution of a followed trajectory of 50 runs. The initial position is randomly sampled from the normal distribution

$$\tilde{\mathbf{p}}_0 \sim \mathcal{N}(\mathbf{p}_0, \Sigma), \quad \Sigma = \begin{pmatrix} 0.1 & 0 & 0 \\ 0 & 0.1 & 0 \\ 0 & 0 & 0.1 \end{pmatrix}. \quad (95)$$

The Figure 21 compares LQR and MPC with prediction horizon $N = 3, N = 10, N = 30$. Emphasizing on the logarithmic y-axis the MPC outperforms the LQR on average by a large margin. Increasing the prediction horizon from three to 30 decreases the error by one order of magnitude.

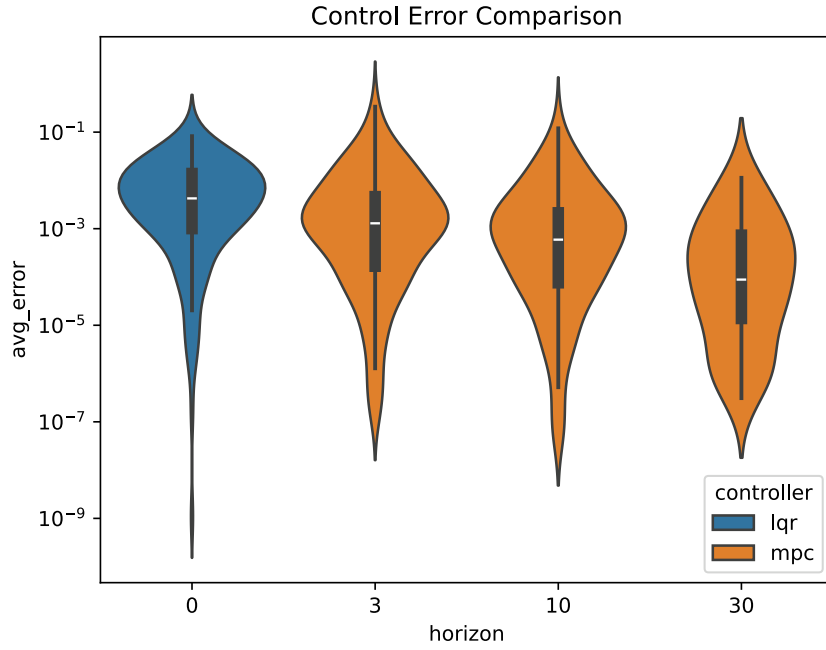


Figure 21: Comparison of average position error of LQR and MPC at different horizon lengths.

Figure 22 reveals that the improved tracking quality comes at the cost of increased compute time. While a horizon of three averages 12.5 seconds of computing. Predicting 30 steps takes 93.6 seconds on average. This linear growth is expected as every prediction step adds 18 decision variables and seven new constraints. As the reference trajectory has a duration of five seconds. The MPC is not real time capable with $N = 3$ in its current implementation. In contrast, the computation time of the LQR as it is only a matrix multiplication is negligible.

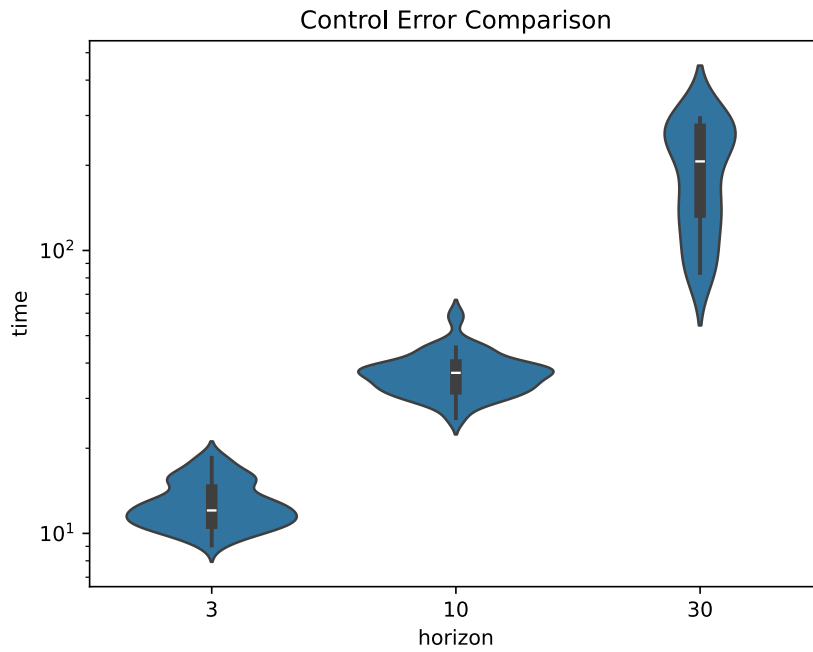


Figure 22: Comparison of compute time of MPC at different horizon lengths.

4.3.5. Multi Robot MPC

The additional constraint for multi robot control was implemented. Unfortunately there are no results due to convergence issues. Increasing the horizon further, experimenting with soft constraints and varying step size could not address the issue.

5. Conclusion

This thesis has explored motion planning and trajectory-following control for hydrofoil vessels, a domain characterized by highly unusual dynamics, a high degree of freedom, strong nonlinearities, and tight constraints on height and attitude to ensure stable flight.

A novel approach to modeling the vessel's dynamics was developed and qualitatively verified using a physical scale model. With this foundation, the capabilities of sampling-based motion planning algorithms, specifically RRT and SST, were investigated for the hydrofoil's unique challenges. The results demonstrated that both algorithms are able to generate feasible trajectories under these special dynamics. RRT proved advantageous when computational resources are limited, offering fast solutions, while SST produced higher-quality paths at the expense of increased computation time. Attempts to refine solutions with optimization-based algorithms such as Gradient Descent did not yield improvements, primarily due to convergence issues.

The second major challenge addressed was the stable following of reference trajectories. Two novel full-state feedback controllers, LQR and MPC, were studied. A symbolic representation of the hydrofoil's dynamics was implemented in a computer algebra system to apply linearization and solve optimization problems. Simulation results showed that LQR provided strong control performance for most states, though it struggled with lateral error. Further analysis revealed that the nonlinear behavior of the hydrofoil vessel's masts introduce linearly uncontrollable dynamics in lateral steering, leading to instability. Nonetheless, LQR offered robust trajectory tracking when lateral control was constrained.

To overcome this limitation, a nonlinear model predictive controller was designed. By employing a sufficiently long prediction horizon of 300 ms, the MPC achieved stable lateral steering and resolved the controllability issue. However, this came at a significant computational cost, rendering the approach unsuitable for real-time implementation in its current form. Shorter horizons mitigated the computational demand but at the expense of lateral stability.

In summary, this work provides new insights into both motion planning and control for hydrofoil vessels. It demonstrates that feasible and efficient trajectories can be planned under complex nonlinear dynamics, and that advanced control strategies such as MPC can address fundamental controllability challenges, albeit with computational trade-offs.

Future Work

While this thesis provides important insights into motion planning and control for hydrofoil vessels, several directions remain open for future exploration.

A first priority is to improve the accuracy of the dynamic model. This includes validation against real-world measurement data, incorporation of actuator rate limits, and explicit consideration of sea-state influences. Moreover, extending the model to capture take-off and landing procedures would broaden its operational relevance. At the same time, selective linearization techniques could be applied to simplify the model, thereby reducing the computational cost of nonlinear MPC while preserving essential dynamics.

In terms of motion planning, future work could investigate reduced-order formulations. Since the vessel's motion is tightly constrained in height, pitch, and roll, while obstacles are confined to the planar XY domain, planning in only three degrees of freedom, potentially using motion primitives or a simplified model, may significantly lower computational requirements. In addition, exploring differential flatness of the hydrofoil dynamics could open the door to efficient optimization-based trajectory generation, similar to methods already established in quadrotor control [50].

The MPC formulation in this thesis is based on reference trajectory following. Recent publications in quad rotor control use model predictive path contouring control. In this approach the Euclidian distance to the reference path is minimized, instead of a specific reference state [51]. This could be applied to hydrofoil dynamics, enabling time optimal path following. Finally, control architectures based on timescale separation offer another promising path of research. The vessel's roll, pitch, yaw, and heave can be regarded as fast dynamics under direct influence, whereas surge and sway represent slower translational dynamics. Splitting the model along these lines would allow cascaded controllers to operate with different sampling rates and prediction horizons, potentially improving performance and computational efficiency [52].

Bibliography

- [1] “The Foiling Revolution.” Accessed: Sep. 27, 2025. [Online]. Available: <https://sailgp.com/news/25/sailgp-betting-the-foiling-revolution/>
- [2] “America's Cup - AC72's Sailing and Foiling.” Accessed: Sep. 27, 2025. [Online]. Available: <https://www.sail-world.com/Australia/Americas-Cup-AC72s-Sailing-and-Foiling/-101667>
- [3] Y. Hosseini, “World's first electric hydrofoil ferry line takes off in Stockholm.” Accessed: Sep. 27, 2025. [Online]. Available: <https://candela.com/newsroom/worlds-first-electric-hydrofoil-ferry-line-takes-off-in-stockholm/>
- [4] “Hydro Motion Boat - the first flying hydrogen-propelled vessel in the world.” Accessed: Sep. 27, 2025. [Online]. Available: <https://www.sail-world.com/news/239084/Hydro-Motion-Boat-first-flying-hydrogen-vessel>
- [5] G. Zogopoulos-Papaliakos, G. C. Karras, and K. J. Kyriakopoulos, “A Fault-Tolerant Control Scheme for Fixed-Wing UAVs with Flight Envelope Awareness,” *Journal of Intelligent & Robotic Systems*, vol. 102, no. 2, p. 46, May 2021, doi: 10.1007/s10846-021-01393-3.
- [6] P. Foehn, A. Romero, and D. Scaramuzza, “Time-Optimal Planning for Quadrotor Waypoint Flight,” *Science Robotics*, vol. 6, no. 56, p. eabh1221, Jul. 2021, doi: 10.1126/scirobotics.abh1221.
- [7] H. Xu, L. Moreira, and C. Guedes Soares, Eds., *Maritime Autonomous Vessels*. MDPI - Multidisciplinary Digital Publishing Institute, 2023. doi: 10.3390/books978-3-0365-6414-2.
- [8] “AI class - Monaco Energy Boat Challenge.” Accessed: Sep. 27, 2025. [Online]. Available: <https://energyboatchallenge.com/challenge/participants/ai-class/>
- [9] J. Hawkins-Dady, J. Hawkins-Dady, and J. Carter, “Foiling Autonomous Surface Vessel: Project C-Flyer,” 2021.
- [10] N. T. Thompson, P. R. Whitworth, and K. I. Matveev, “Development of small-scale unmanned hydrofoil boats,” *Journal of Unmanned Vehicle Systems*, vol. 9, no. 1, pp. 21–32, Mar. 2021, doi: 10.1139/juvs-2019-0019.
- [11] T. I. Fossen, “Handbook of Marine Craft Hydrodynamics and Motion Control,” *Handbook of Marine Craft Hydrodynamics and Motion Control*. John Wiley & Sons, Ltd, p. i–xix, 2011. doi: 10.1002/9781119994138.fmatter.
- [12] Z.-S. Dong, L. Yun, R. Datla, and A. Bliault, *High-Speed Monohull and Hydrofoil Craft: Performance, Technology, and Applications*. Cham: Springer Nature Switzerland, 2025. doi: 10.1007/978-3-031-62760-6.

-
- [13] W. Hönig, “Lecture Flying Robots.” Oct. 2024.
- [14] R. F. Stengel, *Flight Dynamics*. Princeton University Press, 2015. doi: 10.1515/9781400866816.
- [15] P. Liu, *Aerodynamics*. Singapore: Springer Nature, 2022. doi: 10.1007/978-981-19-4586-1.
- [16] M. V. Cook, *Flight dynamics principles: a linear systems approach to aircraft stability and control*, 3rd ed. in Elsevier aerospace engineering series. Waltham, MA: Butterworth-Heinemann, 2012. doi: 10.1016/C2010-0-65889-5.
- [17] J. Paulus, “A Multi-Input Multi-Output Control Model for a Hydrofoil Boat with Differential Front & Rear Strut Steering and Actuated-Wing-Induced Roll,” *Naval Engineers Journal*, vol. 134, no. 1, pp. 33–47, Mar. 2022.
- [18] G. P. van Marrewijk, J. K. Schonebaum, and A. Schwab, “An Experimentally Validated Dynamical Model of a Single-Track Hydrofoil Boat,” *Naval Engineers Journal*, vol. 130, no. 1, pp. 113–134, Mar. 2018.
- [19] R. Bencatel, S. Keerthivarman, I. Kolmanovsky, and A. R. Girard, “Full State Feedback Foiling Control for America’s Cup Catamarans,” *IEEE Transactions on Control Systems Technology*, vol. 29, no. 1, pp. 1–17, Jan. 2021, doi: 10.1109/TCST.2019.2955059.
- [20] J. M. K. Godø and S. Steen, “An efficient method for unsteady hydrofoil simulations, based on Non-Linear Dynamic Lifting Line theory,” *Ocean Engineering*, vol. 288, p. 116001, Nov. 2023, doi: 10.1016/j.oceaneng.2023.116001.
- [21] J. M. K. Godø, S. Steen, and O. M. Faltinsen, “A resistance model for hydrofoil fast ferries with fully submerged foil systems,” *Ocean Engineering*, vol. 301, p. 117503, Jun. 2024, doi: 10.1016/j.oceaneng.2024.117503.
- [22] J. S. Carlton, “Chapter 6 - Propeller Performance Characteristics,” *Marine Propellers and Propulsion (Third Edition)*. Butterworth-Heinemann, Oxford, pp. 79–136, Jan. 2012. doi: 10.1016/B978-0-08-097123-0.00006-X.
- [23] S. M. LaValle, *Planning Algorithms*. Cambridge: Cambridge University Press, 2006. doi: 10.1017/CBO9780511546877.
- [24] M. Pivtoraiko and A. Kelly, “Generating near minimal spanning control sets for constrained motion planning in discrete state spaces,” in *2005 IEEE/RSJ International Conference on Intelligent Robots and Systems*, Aug. 2005, pp. 3231–3237. doi: 10.1109/IROS.2005.1545046.
- [25] M. Elbanhawi and M. Simic, “Sampling-Based Robot Motion Planning: A Review,” *IEEE Access*, vol. 2, pp. 56–77, 2014, doi: 10.1109/ACCESS.2014.2302442.
- [26] L. Kavraki, P. Svestka, J.-C. Latombe, and M. Overmars, “Probabilistic roadmaps for path planning in high-dimensional configuration spaces,”

-
- IEEE Transactions on Robotics and Automation*, vol. 12, no. 4, pp. 566–580, Aug. 1996, doi: 10.1109/70.508439.
- [27] S. LaValle, “Rapidly-exploring random trees : a new tool for path planning,” *The annual research report*, 1998, Accessed: Sep. 03, 2025. [Online]. Available: <https://www.semanticscholar.org/paper/Rapidly-exploring-random-trees-%3A-a-new-tool-for-LaValle/d967d9550f831a8b3f5cb00f8835a4c866da60ad>
 - [28] S. M. LaValle and J. J. Kuffner Jr., “Randomized Kinodynamic Planning,” *The International Journal of Robotics Research*, vol. 20, no. 5, pp. 378–400, May 2001, doi: 10.1177/02783640122067453.
 - [29] “Asymptotically optimal sampling-based kinodynamic planning - Yanbo Li, Zakary Littlefield, Kostas E. Bekris, 2016.” Accessed: Aug. 19, 2025. [Online]. Available: <https://journals.sagepub.com/doi/10.1177/0278364915614386>
 - [30] M. Zucker *et al.*, “CHOMP: Covariant Hamiltonian optimization for motion planning,” *The International Journal of Robotics Research*, vol. 32, no. 9–10, pp. 1164–1193, Aug. 2013, doi: 10.1177/0278364913488805.
 - [31] B. Xing and B. Li, “New Techniques in Motion Control and Path Planning of Marine Vehicles,” *Journal of Marine Science and Engineering*, vol. 12, no. 1, p. 176, Jan. 2024, doi: 10.3390/jmse12010176.
 - [32] R. Dorf and R. Bishop, *Modern Control Systems*. Pearson International, 2021. Accessed: Sep. 03, 2025. [Online]. Available: <https://elibrary.pearson.de/book/99.150005/9781292422350>
 - [33] Y. PEI, S. BISWAS, D. S. FUSSELL, and K. PINGALI, “An Elementary Introduction to Kalman Filtering,” *Communications of the ACM*, vol. 62, no. 11, pp. 122–133, Nov. 2019, doi: 10.1145/3363294.
 - [34] H. Michalska and D. Mayne, “Moving horizon observers and observer-based control,” *IEEE Transactions on Automatic Control*, vol. 40, no. 6, pp. 995–1006, Jun. 1995, doi: 10.1109/9.388677.
 - [35] Z. Ma and S. Zou, *Optimal Control Theory: The Variational Method*. Singapore: Springer, 2021. doi: 10.1007/978-981-33-6292-5.
 - [36] G. Schulz and K. Graf, *Regelungstechnik 2: Mehrgrößenregelung, Digitale Regelungstechnik, Fuzzy-Regelung*. Oldenbourg Wissenschaftsverlag, 2013. doi: 10.1524/9783486736151.
 - [37] J. B. Rawlings, D. Q. Mayne, and M. Diehl, *Model Predictive Control: Theory, Computation, and Design*. Nob Hill Publishing, 2017.
 - [38] S. Gros, M. Zanon, R. Quirynen, A. Bemporad, and M. Diehl, “From linear to nonlinear MPC: bridging the gap via the real-time iteration,” *International Journal of Control*, vol. 93, no. 1, pp. 62–80, Jan. 2020, doi: 10.1080/00207179.2016.1222553.

-
- [39] E. A. Beachy and J. B. Stanhope, "Modification of a Lateral MIMO Model for Use as a Hydrofoil Boat Design Tool and Flight Control System with the Addition of Longitudinal Control," *Naval Engineers Journal*, vol. 134, no. 4, pp. 25–38, Dec. 2022.
- [40] J. v. d. Schaaf, "Using reinforcement learning to control hydrofoils." Accessed: Nov. 27, 2024. [Online]. Available: <http://essay.utwente.nl/89459/>
- [41] J. M. K. Godø, S. Steen, D. Barrett, and M. S. Triantafyllou, "Active wave energy extraction by hydrofoil vessels through deep reinforcement learning-based flight control," *Ocean Engineering*, vol. 312, p. 119236, Nov. 2024, doi: 10.1016/j.oceaneng.2024.119236.
- [42] H. Niu, "The Longitudinal Attitude Control of the Fully-Submerged Hydrofoil Craft with Actuator Saturation," in *2023 42nd Chinese Control Conference (CCC)*, Jul. 2023, pp. 502–506. doi: 10.23919/CCC58697.2023.10241191.
- [43] S. Liu, C. Xu, and L. Zhang, "Hierarchical Robust Path Following Control of Fully Submerged Hydrofoil Vessels," *IEEE Access*, vol. 5, pp. 21472–21487, 2017, doi: 10.1109/ACCESS.2017.2756852.
- [44] S. Helling, M. Lutz, and T. Meurer, "Flatness-based MPC for underactuated surface vessels in confined areas," *IFAC-PapersOnLine*, vol. 53, no. 2, pp. 14686–14691, Jan. 2020, doi: 10.1016/j.ifacol.2020.12.1831.
- [45] G. Terzakis, M. Lourakis, and D. Ait-Boudaoud, "Modified Rodrigues Parameters: An Efficient Representation of Orientation in 3D Vision and Graphics," *Journal of Mathematical Imaging and Vision*, vol. 60, no. 3, pp. 422–442, Mar. 2018, doi: 10.1007/s10851-017-0765-x.
- [46] M. Drela, "XFOIL: An Analysis and Design System for Low Reynolds Number Airfoils," in *Low Reynolds Number Aerodynamics*, T. J. Mueller, Ed., Berlin, Heidelberg: Springer, 1989, pp. 1–12. doi: 10.1007/978-3-642-84010-4_1.
- [47] I. A. Sucan, M. Moll, and L. E. Kavraki, "The Open Motion Planning Library," *IEEE Robotics & Automation Magazine*, vol. 19, no. 4, pp. 72–82, Dec. 2012, doi: 10.1109/MRA.2012.2205651.
- [48] J. A. E. Andersson, J. Gillis, G. Horn, J. B. Rawlings, and M. Diehl, "CasADi: a software framework for nonlinear optimization and optimal control," *Mathematical Programming Computation*, vol. 11, no. 1, pp. 1–36, Mar. 2019, doi: 10.1007/s12532-018-0139-4.
- [49] A. Wächter and L. T. Biegler, "On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming," *Mathematical Programming*, vol. 106, no. 1, pp. 25–57, Mar. 2006, doi: 10.1007/s10107-004-0559-y.

- [50] D. Mellinger and V. Kumar, "Minimum snap trajectory generation and control for quadrotors," in *2011 IEEE International Conference on Robotics and Automation*, May 2011, pp. 2520–2525. doi: 10.1109/ICRA.2011.5980409.
- [51] A. Romero, S. Sun, P. Foehn, and D. Scaramuzza, "Model Predictive Contouring Control for Time-Optimal Quadrotor Flight," *IEEE Transactions on Robotics*, vol. 38, no. 6, pp. 3340–3356, Dec. 2022, doi: 10.1109/TRO.2022.3173711.
- [52] M. A. Sadi, A. Jamali, A. M. N. b. A. Kamaruddin, and V. Y. S. Jun, "Cascade model predictive control for enhancing UAV quadcopter stability and energy efficiency in wind turbulent mangrove forest environment," *e-Prime - Advances in Electrical Engineering, Electronics and Energy*, vol. 10, p. 100836, Dec. 2024, doi: 10.1016/j.prime.2024.100836.

A Notation

A.1 Quaternion Rotation Notation

The models states uses an unit quaterion to represent rotation. There are many conventions to define the rotation opertation of a unit quaterion on a vector. So lets define $\mathbf{q} = (q_w, \mathbf{q}_{i,j,k})$, $\mathbf{q} \in \mathbb{H}$, $q_w \in \mathbb{R}$, $\mathbf{q}_{i,j,k} \in \mathbb{R}^3$ being a unit quaternion representing a rotation in $\mathbb{R}^3$.

Let us define the quaterion multiplication operator $\otimes$ as the hamilton product

$$\mathbf{q} \otimes \mathbf{q}' := \begin{pmatrix} q_w q'_w - q_x q'_x - q_y q'_y - q_z q'_z \\ q_x q'_w + q_w q'_x - q_z q'_y + q_y q'_z \\ q_y q'_w + q_z q'_x + q_w q'_y - q_x q'_z \\ q_z q'_w - q_y q'_x + q_x q'_y + q_w q'_z \end{pmatrix}. \quad (96)$$

To describe the rotation of a vector $\mathbf{v} \in \mathbb{R}$ by quaterion $q \in \mathbb{H}$. We define

$$\mathbf{v}' = \mathbf{q} \odot \mathbf{v} := (\mathbf{q} \otimes (0, \mathbf{v})_{\mathbb{H}} \otimes \mathbf{q}^*)_{i,j,k} \in \mathbb{R}^3 \quad (97)$$

by augmenting $\mathbf{v}$ to a quaterion, and extracting the vector part of the result.

A.2 State Addition

- addition $\oplus : \mathcal{X} \times \mathcal{X} \rightarrow \mathcal{X}$

$$x \oplus x' = \begin{pmatrix} \mathbf{p} + \mathbf{p}' \\ \mathbf{q} \mathbf{q}' \\ \mathbf{v} + \mathbf{v}' \\ \boldsymbol{\omega} + \boldsymbol{\omega}' \end{pmatrix} \quad (98)$$

B Parameters

Parameter	Value	Parameter	Value
$\mathbf{p}_{\text{port,lift}}$	$[1.0, 1.0, -1.0]$	$\mathbf{p}_{\text{starboard,lift}}$	$[-1.0, 1.0, -1.0]$
$\mathbf{p}_{\text{stern,lift}}$	$[-2.0, 0.0, -1.0]$	$\mathbf{p}_{\text{port,mast}}$	$[1.0, 1.0, -0.5]$
$\mathbf{p}_{\text{starboard,mast}}$	$[-1.0, 1.0, -0.5]$	$\mathbf{p}_{\text{stern,mast}}$	$[-2.0, 0.0, -0.5]$
$\mathbf{q}_{\text{port,lift}}$	$[1.0, 0.0, 0.0, 0.0]_{\mathbb{H}}$	$\mathbf{q}_{\text{starboard,lift}}$	$[1.0, 0.0, 0.0, 0.0]_{\mathbb{H}}$
$\mathbf{q}_{\text{stern,lift}}$	$[1.0, 0.0, 0.0, 0.0]_{\mathbb{H}}$	$\mathbf{q}_{\text{port,mast}}$	$[0.71, 0.0, 0.0, 0.71]_{\mathbb{H}}$
$\mathbf{q}_{\text{starboard,mast}}$	$[0.71, 0.0, 0.0, 0.71]_{\mathbb{H}}$	$\mathbf{q}_{\text{stern,mast}}$	$[0.71, 0.0, 0.0, 0.71]_{\mathbb{H}}$
$c_{\text{port,lift}}$	0.15	$c_{\text{starboard,lift}}$	0.15
$c_{\text{stern,lift}}$	0.15	$c_{\text{port,mast}}$	0.15
$c_{\text{starboard,mast}}$	0.15	$c_{\text{stern,mast}}$	0.15
$S_{\text{port,lift}}$	0.15	$S_{\text{starboard,lift}}$	0.15
$S_{\text{stern,lift}}$	0.15	$S_{\text{port,mast}}$	0.15
$S_{\text{starboard,mast}}$	0.15	$S_{\text{stern,mast}}$	0.15
$u_{\text{stb,min}}$	-15	$u_{\text{stb,max}}$	15
$u_{\text{prt,min}}$	-15	$u_{\text{prt,max}}$	15
$u_{\text{stn,min}}$	-15	$u_{\text{stn,max}}$	15
$u_{\text{rud,min}}$	-35	$u_{\text{rud,max}}$	35
m	250	$\mathbf{J}$	$\text{diag}([1000, 1000, 1000])$
$\mathbf{g}$	$[0, 0, -9.81]$	F_{max}	1000
$\mathbf{p}_{\text{thrust}}$	$[-3, 0, -1]$		

Table 3: Model parameters
